The uncomputable price of precision

Seth Douglas Nidhal Mghirbi

Version 2.0.0, October 2026

Abstract

Which numbers can be the entanglement price of perfection for one fixed finite game? Every positive left-computably-enumerable real occurs, and the exponents go strictly beyond this class. For each such α we construct a finite Boolean constraint-system game of quantum value one, with no finite-dimensional perfect strategy, whose least local dimension obeys $\log_2 D_G(\varepsilon) = \alpha \log_2(1/\varepsilon) + o(\log(1/\varepsilon))$. The infimum one-way distillable entanglement of the original mixed strategy state has the same coefficient. A second construction realizes $\alpha = (4 - \theta_K)/4$, a noncomputable right-c.e. real that is not left-c.e., where $\theta_K = \sum_{e \in K} 4^{-(e+1)}$ and K is the halting set. As in our earlier work, every dimension exponent is Δ_2^0 whenever its limit exists. In the reset model of our companion paper on the heat price of precision, local-reset heat has coefficient 2α, while joint-reset heat stays bounded. The halting-degree game of version 1.0.0 and its proofs are retained. A finite address reader, extra clocks and polynomial residual powers give the left-c.e. realization; triangular extraction gives the mixed-state bound. To escape the exact-trace barrier, a fixed polynomial-testing program uses finite perfect witnesses from MIP*=RE, admitted with sublinear logarithmic dimension overhead. Exact and matrix-approximable trace minima then differ. Both left-c.e. and non-left-c.e. uncomputable examples accumulate at zero. The full spectra remain open.

The authors developed this work with current frontier AI models from more than one developer and take responsibility for every claim. How the work was done and checked, including refutation-first proof audits and finite computations, is described before the appendices.

Version 2.0.0. Concept DOI (latest version): 10.5281/zenodo.23701733. The released version 1.0.0 is cited as [11].

Contents

1 The price of perfection 3

2 Weighted models and elementary estimates 6

3 From bounded polynomials to Boolean constraints 7

4 A finite Cantor clock 10

5 The finite programmed presentation 13

6 Every tracial limit pays for halting 15

7 Simultaneous finite models 17

8 Counting measurements on the actual state 19

9 The matched lower bound 21

10 The game and its exponent 25

11 Every positive left-c.e. exponent 26

12 Usable entanglement of the original state 30

13 Heat depends on the order of reset and communication 34

14 The exact-trace barrier and matrix approximation 35

15 Escaping the barrier: a complementary halting flag 38

How this work was done 46

A Why exact periodic verification is insufficient 48

B Finite computations and reproducibility 50

1 The price of perfection

Which numbers can be the entanglement price of perfection for one fixed finite game? The exponent is the rate at which entanglement must grow as a fixed finite device is asked to play closer to perfection; if it can be uncomputable, no algorithm reading the finite rules of the game can predict that cost. Here prediction means computing the limiting exponent. For the entropy of a multidimensional shift of finite type the analogous question has a computability-theoretic answer [16]; here the quantity is the exponent of the least local dimension, and of the usable entanglement, needed to lose at most ε. Every positive left-computably-enumerable real occurs, including uncomputable reals. The exponents also go strictly beyond the left-c.e. reals: $(4-\theta_K)/4$ is a noncomputable right-c.e. example, where $\theta_K = \sum_{e\in K} 4^{-(e+1)}$ and K is the halting set. The same games have their dimension exponent as their logarithmic usable-entanglement coefficient. Their local-reset heat has twice that coefficient, while their joint-reset heat stays bounded.

A real is *left-computably-enumerable*, or *left-c.e.*, if it is the limit of a computable non-decreasing sequence of rationals. Increasing approximations below are understood in this non-strict sense. All logarithms without a displayed base are base two, and entanglement is measured in ebits. For a finite game G of quantum value one, let $D_G(\varepsilon)$ be the least local dimension of a strategy with loss at most ε, padding the two local spaces equally. Define

$$d_G^{\rightarrow}(\varepsilon) = \inf_{\text{finite-dimensional strategies of loss } \le \varepsilon} E_D^{A\to B}(\rho_{AB}). \tag{1.1}$$

The state may be mixed. Its one-way distillable entanglement is the asymptotic Bell-pair rate using local operations and classical communication from Alice to Bob. A real is *right-c.e.* if its negative is left-c.e.; it is Δ_2^0 if computable relative to the halting set.

Theorem 1.1 (Realization)**.** *For every positive left-c.e. real α there is a fixed finite two-player Boolean constraint-system game G_α, with rational question probabilities, quantum value one and no finite-dimensional perfect strategy, such that, as $\varepsilon \downarrow 0$,*

$$\log_2 D_{G_\alpha}(\varepsilon) = \alpha \log_2(1/\varepsilon) + o(\log(1/\varepsilon)), \tag{1.2}$$

$$d_{G_\alpha}^{\rightarrow}(\varepsilon) = \alpha \log_2(1/\varepsilon) + o(\log(1/\varepsilon)). \tag{1.3}$$

There is also such a game $\widehat{G}_K$ satisfying both laws with $\alpha = (4-\theta_K)/4$, which is right-c.e. and not left-c.e. Thus the positive dimension exponents in this BCS class strictly contain the positive left-c.e. reals and are contained in the positive Δ_2^0 reals. Every finite dimension exponent of a rational finite game is Δ_2^0 whenever its defining limit exists.

A finite game's price of precision can have any positive left-c.e. coefficient and can also be non-left-c.e. The coefficient measures usable entanglement in the original mixed state. For the halting examples, local-reset heat has an uncomputable logarithmic coefficient, while joint-reset heat stays bounded. The precise heat statements are Corollary 13.1 and Theorem 15.7.

1.1 The flagship halting game

Fix a standard effective enumeration of programs, with halting set $K \subseteq \mathbb{N} = \{0, 1, \ldots\}$.

Theorem 1.2 (The version 1.0.0 game)**.** *There is a fixed finite two-player Boolean constraint-system game G_K, with rational question probabilities, such that $\omega_q(G_K) = 1$, no finite-dimensional strategy wins perfectly, and*

$$D_{G_K}(\varepsilon) = \varepsilon^{-(3+\theta_K)/4+o(1)}, \qquad \theta_K = \sum_{e\in K} 4^{-(e+1)}. \tag{1.4}$$

The exponent is Turing-equivalent to K. The sequence $\log D_{G_K}(2^{-n})/n$ has no computable certified rate of convergence.

This is the original explicit degree example. The existence of increasing computable rational sequences with uncomputable limits is classical, beginning with Specker [38]; the construction places such a limit inside the resource law of a single finite game. Sections 2–10 retain the construction and proof of [11]. Sections 11–13 extend it. Section 14 proves the exact-trace barrier; Section 15 escapes it and states the remaining classification questions.

1.2 Related work and credit

Mousavi, Nezhadi and Yuen prove that deciding whether the quantum value is exactly one is Π^0_2-complete [27, 28]. In particular, there is no total computable dimension bound, uniform over all value-one games and all precisions. Indeed, such a bound would make value one a universal integer quantification of decidable bounded-dimensional feasibility statements, placing that problem in Π^0_1. Their obstruction is uniform over games. Theorems 1.1 and 1.2 instead identify convergent coefficients for individual fixed games. The theorem $\mathrm{MIP}^* = \mathrm{RE}$ puts halting information in the value of a computable family of games [19]; it does not by itself identify a resource exponent of a fixed value-one game.

The algebraic connection between constraint games and operator assignments goes back to Cleve and Mittal [5] and, for linear systems, Cleve, Liu and Slofstra [4]. Slofstra established an embedding theorem for solution groups [36] and the non-closure of quantum correlations [35]. Slofstra and Vidick introduced the hyperlinear profile as a quantitative bridge from approximate representations to dimension requirements [37]. Slofstra later obtained a finitely presented group with at least subexponential hyperlinear profile [34]. Ji, Leung and Vidick construct a fixed finite three-player game of value one for which near-perfect play requires at least $\Omega(\varepsilon^{-c})$ qubits for a fixed $c > 0$ [18]. Coladangelo constructs a fixed finite two-player score game for which every ε-optimal strategy has local dimension $2^{\Omega(\varepsilon^{-1/8})}$ [6]. These results supply the framework and important growth examples; they do not identify the matched finite exponent in (1.4).

Mehta, Slofstra and Zhao construct halting-dependent Pauli systems and embed them in finite presentations with polynomial bounds on relation decompositions [25, Proposition 4.8]. Those bounds transfer lower estimates; they do not assert small simultaneous finite-model extensions. The construction here supplies its own upper models. The conjugacy gadgets used in our compiler are those of Fanizza, Kroell, Mehta, Paddock, Rochette, Slofstra and Zhao [14, Lemma 5.8 and Theorem 5.5]. Their credit is essential: the passage from nested conjugacy to ordinary Boolean contexts is not a new gadget of this paper. We give its matrices and converse calculation because quantitative completeness is needed here.

The state-preserving Boolean correspondence was established in *The price of precision* [12, Theorem 4.9]; its linear-system predecessor and further dimension examples appear in the companion paper [26]. We use that correspondence as an existing theorem. Section 10 records the elementary square-root estimate needed for the present local-dimension application and checks the constants' effect on the exponent. Theorem 1.1 proves and strengthens the computable-coefficient inclusion conjectured in the discussion of [12, Problem 18.3(i)], for its original usable-entanglement demand as well as dimension. The Δ^0_2 upper bound for dimension exponents is [12, Proposition 11.13]; Lemma 2.4 records the rational case.

Uncomputable asymptotic data in finite systems have other precedents. Hochman and Meyerovitch characterize the entropies of multidimensional shifts of finite type as exactly the nonnegative right-c.e. reals [16]; their construction passes through a shift whose maximal symbol frequency is a prescribed right-c.e. real. Their theorem is the closest template for identifying the possible values of a physical quantity of a fixed finite system with a

computability class. Here every positive left-c.e. real is realized, together with a non-left-c.e. right-c.e. family; equality for the full game spectrum is not asserted. Sapir, Birget and Rips relate group isoperimetry and computation [32]; a Dehn exponent is not a dimension exponent, and no identification between them is used here. Cubitt, Perez-Garcia and Wolf prove undecidability of the spectral gap over families of local Hamiltonians [8]. Bausch, Cubitt and Watson obtain uncomputable phase diagrams for a continuous one-parameter family [1]. These are precedents for asymptotic noncomputability, rather than proofs of (1.4).

The binary adding machine used below is the classical odometer on a Cantor set; its interpretation through operator algebras is standard [15]. The new point needed here is a finite polynomial presentation and a carry estimate that remain quantitative in a non-flat state. The final information bound uses Holevo's theorem [17] and the random-access-code principle exemplified by Nayak [30]. Its branching estimate is proved explicitly.

The density classifications of Jockusch–Schupp and Downey–Jockusch–Schupp [20, 13] are prior work on computation encoded in densities. The moving-window estimate in Section 11 uses the binary radical-inverse ordering of van der Corput [39], not a new discrepancy principle. Vidick's *Almost synchronous quantum correlations* [40] supplies the closely related spectral-compression mechanism; Section 14 gives the word-moment form needed here. Tracial moment optimization [2, 22] supplies the exact-trace lower hierarchy. Its distinction from general commuting-model NPA [29] is the distinction used to escape the barrier in Section 15.

Devetak–Winter hashing [9] is the operational last step in Section 12. The new ingredient is the inactive-outcome extension of the triangular extraction in [12, Theorem 6.2] and its cyclic finite-alphabet form [12, Lemma 6.3], with random targets and an original-mixed-state bridge. The heat conclusion invokes the existing reset sandwich of [10, Theorem 2.1 and Corollary 2.2]. Kolchinsky and Wolpert [23] study noncomputable heat functions for universal Turing machines. Corollary 13.1 and Theorem 15.7 identify uncomputable logarithmic precision coefficients for fixed finite nonlocal games and compare their local and joint reset schedules. The halting-degree dimension construction, its compiler, clocks and proofs appeared in version 1.0.0 [11]; they are retained here. The new realization, extraction and consequences are proved in the added sections. Section 15 additionally uses the finite perfect-completeness form of Ji–Natarajan–Vidick–Wright–Yuen, states its exact theorem numbers, and credits the Connes embedding and commuting-complexity context, including Lin [24].

1.3 Proof outline

At source precision 4^{-N}, three compulsory clocks cost $3N-o(N)$ bits of logarithmic dimension. They hold an address, fuel and time. For G_K, a finite counter program recognizes halting inputs on separated base-four cylinders. Reset forces initialized computations at every finite fuel scale. Exact tracial models must flag every halting cylinder; compactness extends this to all weighted approximate models, without an effective modulus.

A conditional clock contributes $\theta_K N - o(N)$ further bits. Explicit finite histories are diagonal labels on the existing registers. A moving register of optional bits realizes the conditional clock in dimension $2^{\theta_N N+O(\log N)}$, with computable $\theta_N \uparrow \theta_K$. Source precision 4^{-N} becomes game loss of order 16^{-N}, giving the denominator four.

For an arbitrary left-c.e. $\theta \in [0,1]$, an address reader accepts exactly the binary values below θ. Radical-inverse window counts match its finite flag density. Extra compulsory clocks add to the numerator, and replacing polynomial residuals by fixed powers divides the exponent. These two operations cover every positive left-c.e. α. Finally a coherent extraction with an inactive outcome converts the expected active count into distillable Bell pairs on the original mixed state. The cited heat sandwich then matches both reset laws.

The complementary construction reverses the flag rule: an unflagged decoded cylinder must carry a finite perfect witness for its halting game. A fixed counter transducer with two

operator registers tests every relation of that witness's projection presentation, packed into three data generators. Nonhalting unflagged corners have zero weight in matrix-approximable traces, and moment compression transfers that bound to the actual strategy state. Finite perfect witnesses can be admitted so slowly that their combined logarithmic dimension is at most $\sqrt{N}$, while the unflagged density tends to θ_K. The flag density is therefore $1-\theta_K$, giving exponent $(4-\theta_K)/4$. The same extraction and heat sandwich apply. The exact-trace minimum is strictly smaller, as the barrier forces it to be.

2 Weighted models and elementary estimates

Definition 2.1. A bounded polynomial presentation consists of finitely many variables, each declared to be a positive contraction, a projection, a self-adjoint unitary, or a unitary, and finitely many rational real-coefficient noncommutative $*$-polynomial equations $r = 0$. Its weighted precision-δ model on $\mathbb{C}^d$ is an assignment in the declared domains and a density matrix ρ such that, with $\xi = \sqrt{\rho}$,

$$\|r\xi\|_2 \le \delta, \qquad \|[s,\xi]\|_2 \le \delta \tag{2.1}$$

for each defining relation r and each variable s. The norm is the unnormalized Hilbert–Schmidt norm. A flat model has $\rho = I/d$ and $\|r\|_f = d^{-1/2}\|r\|_2$. Write $W(\delta)$ and $H(\delta)$ for the least weighted and flat model dimensions, with value ∞ if no model exists.

We write $|D|_\rho = \|D\xi\|_2$. Constants for a fixed presentation may depend on its finite polynomials, but never on d, ρ, δ or a growing digit depth.

Lemma 2.2 (Right multiplication)**.** *For bounded A, B,*

$$|AB|_\rho \le \|A\||B|_\rho, \qquad |BA|_\rho \le \|A\||B|_\rho + \|B\|\|[A,\xi]\|_2.$$

A word w of length l in contraction variables and unitary adjoints satisfies $\|[w,\xi]\|_2 \le l\delta$. A fixed finite algebraic derivation from the defining relations has error at most $C\delta$ on the same state.

Proof. Use $BA\xi = B\xi A + B[A,\xi]$. Expand the commutator of a product one factor at a time. Adjoint unitaries have the same state-commutator norm as the unitary. For adjoint relations,

$$|r^*|_\rho = \|\xi r\|_2 \le |r|_\rho + \|[r,\xi]\|_2.$$

All variables are bounded, so a fixed polynomial has a fixed operator-norm bound and a commutator bound $C\delta$. Expand a finite ideal identity as a finite sum $a_j r_j b_j$ and apply the first two inequalities. This proves the last assertion. □

Lemma 2.3 (Hilbert–Schmidt functional calculus)**.** *Let A, B be self-adjoint, Z an arbitrary rectangular Hilbert–Schmidt matrix, and g an L-Lipschitz function on their spectra. Then*

$$\|g(A)Z - Zg(B)\|_2 \le L\|AZ - ZB\|_2, \tag{2.2}$$

$$|g(A) - g(B)|_\rho \le L\big(|A-B|_\rho + 2\|[B,\xi]\|_2\big), \tag{2.3}$$

$$|[Y, g(A)]|_\rho \le L\big(|[Y,A]|_\rho + 2\|[A,\xi]\|_2\big) \tag{2.4}$$

in the last line for a contraction Y. Also $\|[g(A),\xi]\|_2 \le L\|[A,\xi]\|_2$.

Proof. Diagonalize A on the left and B on the right. Corresponding entries in the first inequality are $(g(a) - g(b))Z_{ab}$ and $(a - b)Z_{ab}$. Square the scalar Lipschitz inequality and sum. The state-commutator estimate is its case $A = B, Z = \xi$. Insert and subtract $\xi g(B)$ to obtain (2.3). For (2.4), apply (2.2) with $A = B, Z = Y\xi$; then insert and subtract $Y\xi g(A)$. The two resulting state-commutator terms are each bounded by $L\|[A, \xi]\|_2$. These are Schatten-two estimates, not operator-norm Lipschitz statements. □

Lemma 2.4 (Finite-precision computability)**.** *For a rational-data game promised to have quantum value one, $D_G(\varepsilon)$ is computable at every positive rational ε. If $a_n = \log D_G(2^{-n})/n$ converges to β and $|a_n - \beta| \le r(n)$ eventually for a computable $r(n) \to 0$, then β is computable.*

Proof. In a fixed local dimension d, introduce real variables for the real and imaginary matrix entries of the state and both families of POVMs. Positivity is a finite list of principal-minor inequalities. Normalization and winning probability at least $1 - \varepsilon$ are polynomial conditions with rational coefficients. Real quantifier elimination decides feasibility. Test $d = 1, 2, \ldots$; the value-one promise ensures termination. For the second assertion, hardcode a threshold beyond which the error bound holds. Given k, search above it for n with $r(n) < 2^{-k-1}$ and compute a_n within 2^{-k-1}. This gives a 2^{-k} approximation to β. □

3 From bounded polynomials to Boolean constraints

A BCS specifies finitely many Boolean variables and finitely many contexts. A context lists variables and its allowed sign assignments. In an operator model the context variables are commuting reflections and their joint spectrum is restricted to those assignments. Its game asks Alice for a valid context assignment and Bob for one sign, testing consistency on an incidence. We use the uniform distribution on the finite incidence set, hence rational probabilities. The algebraic interpretation is standard [5, 4].

Theorem 3.1 (Polynomial compiler)**.** *For each finite bounded polynomial presentation $\mathcal{P}$, there are a finite ordinary BCS $\mathcal{B}$ and constants $c, C > 0$ with the following properties.*

(a) Every weighted precision-η model of $\mathcal{B}$ reconstructs a precision-$C\eta$ model of $\mathcal{P}$ on the same space and with the same state.

(b) Every flat precision-δ model of $\mathcal{P}$ on $\mathbb{C}^d$ gives a strategy for its context-variable game in local dimension at most cd, with loss at most $C\delta^2$.

All constants depend only on the presentation. In particular, guarded equations such as $(V - W)(I - A) = 0$ are compiled with their guard intact.

3.1 Domains and selectors

Every positive contraction is

$$r = \frac{2I + u + u^*}{4}, \qquad u = 2r - I + 2i\sqrt{r(I - r)}. \tag{3.1}$$

The real and imaginary parts in this expression commute and their squared sum is I, so u is unitary. Conversely the first formula is a positive contraction for any unitary. For upper models realify the whole assignment by

$$\mathcal{R}(M) = \begin{pmatrix} \Re M & -\Im M \\ \Im M & \Re M \end{pmatrix}.$$

This doubles dimension, preserves real-coefficient equations and flat norms, and takes unitaries to real orthogonal matrices. A real orthogonal matrix is a product of two symmetric orthogonal involutions: on a rotation plane use

$$\begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \cos t & -\sin t \\ -\sin t & -\cos t \end{pmatrix},$$

and use scalar blocks on the remaining eigenspaces. Orthogonal conjugation and direct sums finish the factorization. Replace each unitary by a product of two reflections. Lower reconstructions are automatically unitary, and (3.1) is automatically positive. Their state commutators cost a fixed factor. No commutation between the two factors is imposed. The upper models are flat, so no state estimate for the square root in (3.1) is needed.

A source equation is now $p = \sum_j a_j W_j = 0$, with rational a_j and reflection words W_j. Clear denominators with an integer c, duplicate signed terms, and add cancelling pairs until

$$\sum_{j=1}^{L} \widetilde{W}_j = 2cp, \qquad L = 2^q.$$

Introduce q Pauli pairs X_i, Z_i, commuting across pairs and with every source reflection. Let e_j be the joint Z spectral projections and E the joint $+1$ projection of the X_i. Then

$$Ee_jE = L^{-1}E, \qquad \sum_{v\in\{0,1\}^q} Z^v E Z^v = I. \tag{3.2}$$

For the first identity, expand e_j in Z monomials. Every nonconstant monomial anticommutes with some X_i and hence has zero compression by E. The second identity sums the joint X spectral projections.

The controlled reflection $(I - e_j) + e_j s$ is a Boolean truth table in commuting variables. Products of these controls and controlled signs form a reflection word whose value is the unitary

$$S = \sum_j e_j \widetilde{W}_j.$$

The projection E receives its own Boolean variable x_E. Its defining context is $\{X_1, \dots, X_q, x_E\}$, whose allowed assignments are those with $x_E = +1$ exactly when $X_1 = \cdots = X_q = +1$. In an operator model this truth table says $(I + x_E)/2 = \prod_i (I + X_i)/2 = E$. Introduce $x_F = S x_E S^*$ by successive reflection conjugacies and put $F = (I + x_F)/2 = SES^*$. Orthogonality is imposed on this pair alone: the context $\{x_E, x_F\}$ forbids exactly the assignment $(+1, +1)$, which in an operator model is the condition $EF = 0$. The exact identities are

$$EFS = ESE = \frac{2c}{L} pE, \qquad p = \frac{L}{2c} \sum_v Z^v EFS Z^v. \tag{3.3}$$

They show why a low-weight selector corner cannot conceal an equation. All of its translates are used.

3.2 The conjugacy gadgets

We use the gadgets of [14, Lemma 5.8]. Their formulas are included to establish the precise finite-dimensional extension needed here. For a general relation $x_1 x_2 x_1 = x_3$, introduce commuting parity contexts

$$x_i y_i z_i = I, \qquad x_i f w_i = I, \qquad g y_2 z_3 = I,$$

and the conjugacies $fy_if = z_i$ and $w_1y_2w_1 = z_3$. The targets of these remaining conjugacies commute by the parity contexts. If the original reflections are X_i, an exact doubled extension is

$$x_i = \mathrm{diag}(X_i, X_i), \quad y_i = \mathrm{diag}(X_i, I), \quad z_i = \mathrm{diag}(I, X_i),$$
$$f = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}, \quad w_i = \begin{pmatrix} 0 & X_i \\ X_i & 0 \end{pmatrix}, \quad g = \mathrm{diag}(X_2, X_3).$$

Conversely $w_1 = x_1 f$, $[f, x_1] = 0$, and $fy_2f = z_2$ give $x_1z_2x_1 = z_3$. Conjugation by f gives $x_1y_2x_1 = y_3$. Multiplying gives the required conjugacy.

For $aba = c$ with commuting targets b, c, use six commuting parity contexts:

$$ad_1d_2 = bd_2d_3 = d_3d_4d_5 = ad_5d_6 = cd_6d_7 = d_1d_4d_7 = I. \tag{3.4}$$

They imply

$$aba = d_1d_2d_2d_3d_5d_6 = d_1d_4d_6 = d_7d_6 = c.$$

Writing $M(a) = \left(\begin{smallmatrix} 0 & a \\ a & 0 \end{smallmatrix}\right)$, an exact extension, with old variables repeated diagonally, is

$$d_1 = M(a), \quad d_2 = M(I), \quad d_3 = M(b), \quad d_4 = \begin{pmatrix} 0 & ba \\ ab & 0 \end{pmatrix},$$
$$d_5 = \mathrm{diag}(bab, a), \quad d_6 = \mathrm{diag}(bc, I), \quad d_7 = \mathrm{diag}(b, c).$$

For clarity, $d_4^* = d_4$ and $d_4^2 = I$. The equality $aba = c$ implies that a commutes with bc, while bc is a reflection. These facts verify the fourth context; the other five follow by direct multiplication. Each listed triple consists of commuting reflections with product I. Sequentially applying the two gadgets has constant dimension cost for a fixed presentation. Pauli anticommutation is $XZX = -Z$, with a Boolean variable for the negation of Z.

3.3 Weighted soundness and flat completeness

Proof of Theorem 3.1. All identities above are finite algebraic calculations in the defining relations. A Boolean truth table gives finite algebraic relations: expand in its assignment projections, and any polynomial vanishing on allowed assignments is the finite sum of its values on forbidden assignments times their projections. Thus the controlled-variable definitions, both gadget converses and (3.3) are finite ideal derivations. In an approximate model put $E_{\mathrm{pol}} = \prod_i (I + X_i)/2$, $E_0 = (I + x_E)/2$ and $F_0 = (I + x_F)/2$. The defining AND context of x_E and the conjugacy gadgets give $|E_{\mathrm{pol}} - E_0|_\rho = O(\eta)$ and $|SE_{\mathrm{pol}}S^* - F_0|_\rho = O(\eta)$, while the orthogonality context gives $|E_0F_0|_\rho = O(\eta)$. Interpret (3.3) as a polynomial identity modulo the Pauli and controlled-variable relations. Substitution of these estimates, using Lemma 2.2, gives $|p|_\rho = O(\eta)$. Only E_0 and F_0 are automatically projections here; the selector polynomial need not be a projection in an approximate model. Lemma 2.2 proves linear weighted soundness. This statement is stronger than mere exact implication; the explicit finite derivations are what justify the linear error.

For upper models choose the selectors on an exact tensor factor and define every controlled variable and successive conjugate projection canonically. Put $x_E = 2E - I$ with $E = \prod_i (I + X_i)/2$. It commutes with every X_i, and the joint spectrum of $(X_1, \ldots, X_q, x_E)$ lies in the allowed assignments, so the defining context of x_E is exact. The conjugacies defining x_F are exact even when p is nonzero. All gadget contexts extend exactly. The only inexact constraint is the final orthogonality, where

$$\|EF\|_f = \|ESE\|_f = \frac{2c}{L\sqrt{L}}\|p\|_f. \tag{3.5}$$

Let F' be the projection onto $(I-E)\operatorname{ran} F$. Then

$$EF' = 0, \qquad \|F - F'\|_f^2 \le 2\|EF\|_f^2. \tag{3.6}$$

To prove this, diagonalize FEF on $\operatorname{ran} F$ with eigenvalues λ_i and orthonormal eigenvectors v_i. For $\lambda_i < 1$, the vectors $(I-E)v_i/\sqrt{1-\lambda_i}$ are orthonormal and span the new range. Those with $\lambda_i = 1$ disappear. Hence

$$\|F - F'\|_2^2 = \#\{i : \lambda_i = 1\} + 2\sum_{\lambda_i<1} \lambda_i \le 2\sum_i \lambda_i = 2\|EF\|_2^2.$$

On the orthogonality context $\{x_E, x_F\}$, Alice answers with the PVM

$$E \mapsto (+1,-1), \qquad F' \mapsto (-1,+1), \qquad I - E - F' \mapsto (-1,-1).$$

These are orthogonal projections summing to I, because $F' \le I - E$, and every outcome is an allowed assignment. Her operator for x_E is therefore $2E - I$ and her operator for x_F is $2F' - I$. Bob uses $2E - I$ and $2F - I$. On a maximally entangled state there is no disagreement for x_E, and the disagreement probability for x_F is $\|F - F'\|_f^2$. This is why E carries its own variable: if the context listed $X_1, \dots, X_q$ together with x_F, the PVM above would not be a legal answer, since F' commutes with E but in general not with each X_i, and so does not fix their values. All other contexts, including the defining context of x_E, are exact. With incidence probability π, summing over the finitely many source equations and using (3.6) and (3.5) bounds the loss by

$$2\pi \sum \|EF\|_f^2 = 8\pi \sum \frac{c^2}{L^3} \|p\|_f^2 \le C\delta^2.$$

Here c and L are the integers of each source equation. The new variable adds no tensor factor. Realification, selectors and gadget extensions multiply the dimension by the fixed factor $2^{1+q_{\mathrm{tot}}+g}$, where q_{tot} is the total number of selector qubits and g the number of gadget doublings; this is the dimension constant in (b). This proves both assertions. □

4 A finite Cantor clock

The clock has a positive contraction A, a projection P, a unitary T and a self-adjoint unitary X, with relations

$$[A, P] = 0, \quad TAT^* = 4A - 3P, \quad XPX = I - P, \quad [X, TAT^*] = 0. \tag{4.1}$$

For $j \in \mathbb{N}$ put $P_j = T^j P T^{-j}$ and $X_j = T^j X T^{-j}$. The Cantor set used here is

$$\mathcal{K} = \left\{ \sum_{j\ge 0} 3b_j 4^{-j-1} : b_j \in \{0,1\} \right\}.$$

Let p be its first-digit function, extended to $[0,1]$ by

$$p(a) = \begin{cases} 0 & a \le 1/4, \\ 2a - 1/2 & 1/4 \le a \le 3/4, \\ 1 & a \ge 3/4, \end{cases} \qquad s(a) = 4a - 3p(a).$$

These functions are respectively 2- and 4-Lipschitz and map $[0,1]$ into itself.

Lemma 4.1 (Exact digits)**.** *In an exact model, A has spectrum in $\mathcal{K}$, all P_j are commuting spectral functions of A, and*

$$A = \sum_{j<M} 3 \cdot 4^{-j-1} P_j + 4^{-M} T^M A T^{-M}. \tag{4.2}$$

The flip X_j exchanges P_j and $I - P_j$ and preserves all P_l with $l > j$. In every tracial state,

$$\tau\left(\prod_{j<M} P_j^{b_j} (I - P_j)^{1-b_j}\right) = 2^{-M}. \tag{4.3}$$

The spectral projection of A at 1 is zero in the faithful tracial representation.

Proof. Since A, P commute and $0 \le 4A - 3P \le I$, the $I - P$ block of A has spectrum in $[0, 1/4]$ and the P block in $[3/4, 1]$. Hence $P = p(A)$ and $TAT^* = s(A)$. Conjugating this statement repeatedly proves $P_j = p(s^j(A))$ and (4.2). Its remainder has norm at most 4^{-M}, which places the spectrum in $\mathcal{K}$. The tail coordinate TAT^* determines every later digit. Its commutation with X, and $XPX = I - P$, give the flip assertions, then their translates. A trace assigns equal mass to a cylinder and its earliest-bit flip. Add the two and repeat on the remaining suffix to obtain (4.3). The spectral projection at 1 lies under every all-ones cylinder and has trace at most 2^{-M} for every M. Faithfulness makes it zero. □

Lemma 4.2 (Linear spectral repair)**.** *There is an absolute constant C such that any weighted precision-δ clock has a self-adjoint $\widehat{A}$ with spectrum in $\mathcal{K}$ satisfying*

$$|\widehat{A} - A|_\rho + \|[\widehat{A}, \xi]\|_2 \le C\delta, \tag{4.4}$$

$$|P_j - p(s^j(\widehat{A}))|_\rho \le C 4^j \delta. \tag{4.5}$$

The first M raw digit projections and triangular flips satisfy digit–digit commutations, own-bit flips, later-digit preservation and state commutations with error at most $C(M+1)^3 4^M \delta$. The spectral digit projections $p(s^j(\widehat{A}))$ commute exactly and differ from the raw ones by (4.5).

Proof. Set $B_0 = TAT^*$, $D_0 = PB_0P + (I-P)B_0(I-P)$ and $C_0 = (D_0 + 3P)/4$. The affine relation and $[A, P]$ give $|[B_0, P]|_\rho = O(\delta)$ by Lemma 2.2. Since $B_0 - D_0 = (2P - I)[P, B_0]$, it follows that $|D_0 - B_0|_\rho = O(\delta)$ and $|C_0 - A|_\rho = O(\delta)$. Their state commutators are $O(\delta)$ from their fixed polynomial definitions. All are positive contractions, and D_0 commutes with P.

Let $d_\mathcal{K}$ be distance to $\mathcal{K}$, a 1-Lipschitz function. Cantor self-similarity on the two P blocks gives $d_\mathcal{K}(C_0) = d_\mathcal{K}(D_0)/4$. Lemma 2.3 and unitary transport give

$$|d_\mathcal{K}(A)|_\rho \le \tfrac{1}{4} |d_\mathcal{K}(A)|_\rho + O(\delta).$$

Thus $|d_\mathcal{K}(A)|_\rho = O(\delta)$. Round each spectral value of A to a nearest point of $\mathcal{K}$. The resulting $\widehat{A}$ has $|\widehat{A} - A|_\rho = O(\delta)$. For any self-adjoint E, $\|[E, \xi]\|_2 \le 2|E|_\rho$; apply this to $E = \widehat{A} - A$ to prove (4.4). No continuity of the nearest-point map is needed.

Since $p(C_0) = P$, functional calculus gives $|P - p(\widehat{A})|_\rho = O(\delta)$, and then the affine relation gives $|T\widehat{A}T^* - s(\widehat{A})|_\rho = O(\delta)$. For $g = p \circ s^r$, the one-step error

$$|Tg(\widehat{A})T^* - g(s(\widehat{A}))|_\rho \le C 4^r \delta$$

follows from (2.3). Telescope these identities to depth j. The sum is $O(4^j\delta)$. Every digit function has norm at most one, so transporting the errors through the outer powers of T adds only $O(j^2\delta)$, absorbed by $O(4^j\delta)$. This proves (4.5).

For the flips, translate $XPX = I - P$ and replace each P_j using (4.5). For $l > j$, use the tail coordinate $T^{j+1}AT^{-(j+1)}$. The translated last relation of (4.1) makes its commutator with X_j at most $C(j+1)\delta$ in weighted norm. Its digit at position $l-j-1$ is a function with Lipschitz constant at most $2 \cdot 4^{l-j-1}$. Apply (2.4), replace the digit back, and use Lemma 2.2. The error is at most $C(M+1)^3 4^M \delta$. Raw digit commutators follow by replacing both with commuting functions of $\widehat{A}$; the two replacement costs add. Finally $\|[P_j, \xi]\|_2, \|[X_j, \xi]\|_2 \le (2j+1)\delta$. This proves all assertions. □

Remark 4.3. The spectral repair and digit estimate use only positivity, the affine equation, $[A, P] = 0$, and state commutators. They apply without a flip. This will be used for the conditional coordinates.

Theorem 4.4 (Clock profile)**.** *For the presentation* (4.1)*,*

$$\log W(2^{-n}) = \frac{n}{2} + O(\log(n+2)), \qquad \log H(2^{-n}) = \frac{n}{2} + O(\log(n+2)).$$

There are flat models on N qubits of precision $3 \cdot 4^{-N}$.

Proof. On N qubits let P_j read bit j, let T rotate contents from slot j to slot $j+1$ cyclically, and let X flip bit zero. Put

$$A_N = \sum_{j=0}^{N-1} 3 \cdot 4^{-j-1} P_j. \tag{4.6}$$

Then $[A_N, P_0] = 0$, $XP_0X = I - P_0$, and

$$TA_NT^* - 4A_N + 3P_0 = 3 \cdot 4^{-N} P_0, \qquad [X, TA_NT^*] = 3 \cdot 4^{-N} [X, P_0].$$

All state commutators vanish in the flat state, proving the upper bound. For the lower bound, use M digits and flips from Lemma 4.2 in the binary case of the counting lemma, Theorem 8.1, proved below. Choose $M = \lfloor \log_4(1/\delta) - 20 \log_4(2 + \log_4(1/\delta)) \rfloor$. Its error terms tend to zero, so $\log d \ge M - o(1)$. At $\delta = 2^{-n}$ this is $n/2 - O(\log(n+2))$. Since $W \le H$, both conclusions follow. This forward reference uses no property of the program constructed later. □

4.1 Guarded increment

Add a unitary V and the relation

$$(V - X[(I-P) + (TVT^*)P])(I - A) = 0. \tag{4.7}$$

Proposition 4.5. *The finite clocks have increment models of dimension 2^N whose guarded defect has operator norm $\sqrt{2}\, 4^{-N}$ and flat norm $\sqrt{2}\, 4^{-N} 2^{-N/2}$. In an exact faithful tracial model, V increments every finite binary digit prefix.*

Proof. Let V_N add one modulo 2^N, with bit zero least significant. If that bit is zero, the right-hand side before the guard is the correct first-bit flip. If it is one and some later bit is zero, TV_NT^* carries through slots $1, 2, \ldots$ to the first zero and X then clears slot zero. Only the all-ones input fails:

$$R_N := V_N - X[(I - P_0) + (TV_NT^*)P_0] = (|0^N\rangle - |10^{N-1}\rangle)\langle 1^N|.$$

The guard equals 4^{-N} there. The two norm statements follow from the rank-one formula. For exact tracial models, Lemma 4.1 makes $I - A$ have dense range, so its bounded left factor in (4.7) vanishes. Put $C = TVT^*$ and $D = (I-P) + CP$. Since $V = XD$, D is unitary. It fixes $\ker P$ pointwise, so unitarity forces $\operatorname{ran} P$ onto itself; there D is C. Hence $[C, P] = 0$. The recursion now reads binary increment on the first bit followed by increment on the tail on its carry branch. Induction on prefix length proves the assertion. □

Proposition 4.6 (Mass near the vanishing guard)**.** *There is an absolute constant C such that every weighted precision-δ clock and every integer $M \geq 1$ satisfy*

$$\mathrm{Tr}(\rho\, 1_{[1-4^{-M},1]}(A)) \leq 2^{-M} + C(M+1)^5 4^M \delta. \tag{4.8}$$

If $\|R\| \leq L$ and $|R(I-A)|_\rho \leq \delta$, then

$$|R|_\rho \leq 4^M(1+L)\delta + L\sqrt{2^{-M} + C(M+1)^5 4^M \delta}. \tag{4.9}$$

In particular $|R|_\rho = O((1+\log(1/\delta))^{5/2}\delta^{1/6})$ for fixed L.

Proof. Here it is useful to avoid rounding A first. From the proof of Lemma 4.2, $|P-p(A)|_\rho = O(\delta)$ and $|TAT^* - s(A)|_\rho = O(\delta)$. The same telescoping gives $|P_j - p(s^j(A))|_\rho \leq C4^j\delta$. Threshold $p(s^j(A))$ at $1/2$ to commuting projections R_j. Against an exact projection the cross-spectrum Lipschitz bound for this threshold is two: for $b \in \{0,1\}$ and $a \in [0,1]$, $|1_{[1/2,1]}(a) - b| \leq 2|a-b|$. The eigenbasis proof of Lemma 2.3, followed by its state-commutator insertion, therefore gives $|R_j - P_j|_\rho \leq C(j+1)4^j\delta$. Their commutators with ξ and the triangular flip defects are at most $\eta = C(M+1)^3 4^M \delta$.

For completeness, an approximately invariant suffix law differs from its earliest-bit swap by at most $CM^2\eta$. Pinch ξ in the commuting R_j to ζ; each pinching costs at most η, so $\|\xi - \zeta\|_2 \leq M\eta$. Intertwining a flip with one recorded digit has error $O((M+1)\eta)$ on ζ. Inserting previous joint spectral projections does not enlarge the sum of squared branch errors, since they commute with ζ. Telescoping at most M digits and returning to ξ costs $CM^2\eta$. The initial state changes by trace norm at most 2η under its flip. Purification distance now bounds the claimed total variation.

Let q_i be the probability that $R_i, \ldots, R_{M-1}$ are all one, with $q_M = 1$. The swap bound gives $|2q_i - q_{i+1}| \leq CM^2\eta$, hence $q_0 \leq 2^{-M} + CM^2\eta$. If $a \geq 1 - 4^{-M}$, then $s^j(a) = 1 - 4^j(1-a) \geq 3/4$ for $j < M$. Thus all R_j are one on that spectral interval. This proves (4.8).

Put $G = I - A$ and $E = 1_{[0,4^{-M}]}(G)$. Since $R\xi G = RG\xi + R[\xi, G]$, its norm is at most $(1+L)\delta$. On $I - E$ the inverse of G has norm at most 4^M, while $\|R\xi E\|_2 \leq L\sqrt{\mathrm{Tr}(\rho E)}$. Add the two column bounds and use (4.8). Finally choose $M = \lfloor \frac{1}{3}\log_2(1/\delta) \rfloor$. Both 2^{-M} and $4^M\delta$ are of order $\delta^{1/3}$, proving the last assertion. The relaxed polynomial factor here is harmless; only exact guard cancellation is needed in the flag proof. □

5 The finite programmed presentation

Fix a finite deterministic counter program recognizing K. Counters take nonnegative integer values and instructions increment, decrement on a nonzero branch, or test zero. Such a recognizer follows from the classical universality of counter machines [33]. One direct construction stores the two halves of a Turing tape as integer-coded stacks. Pushing multiplies a stack code by a fixed base and adds a digit; popping computes quotient and remainder by that base. Increment, decrement and zero-test loops perform these operations using finitely many scratch counters. A unary input loop writes the integer input on the simulated tape. This gives one fixed finite instruction table. Add absorbing idle and halt controls.

The variables introduced below have the domains stated explicitly. Projection inclusions $Q \leq F$ mean the polynomial equation $(I-F)Q = 0$. A precision model satisfies every listed commutator and every displayed defining equation to precision δ in Definition 2.1.

5.1 Three compulsory clocks and a carry coordinate

Use three copies of (4.1), with indices a, f, t for address, fuel and time. Complete variable sets of different clocks commute. Add an address increment V and a time increment U,

satisfying (4.7) for their own clocks. The address and fuel sets commute with U, and the time and fuel sets commute with V. Add a positive contraction H with

$$H = (I - P_a) + \tfrac{1}{4} P_a T_a H T_a^*, \tag{5.1}$$

$$[P_a, T_a H T_a^*] = [H, A_a] = 0, \qquad V^* A_a V = A_a - I + \tfrac{7}{4} H. \tag{5.2}$$

Include H in the address set, commuting with all fuel and time variables. The last equality will control many address translations without an exponential precision cost.

5.2 Address decoding and reset

Put $P_{a,1} = T_a P_a T_a^*$, $E_{00} = (I - P_a)(I - P_{a,1})$ and $E_1 = P_a(I - P_{a,1})$. Add a projection L and a positive contraction R with

$$L = E_1 + E_{00} T_a^2 L T_a^{-2}, \tag{5.3}$$

$$R = (I - E_{00}) + \tfrac{1}{2} E_{00} T_a^2 R T_a^{-2}. \tag{5.4}$$

The products are polynomials even before the approximate digits commute. In exact models the cylinder

$$B_e = \left(\prod_{j<e} T_a^{2j} E_{00} T_a^{-2j} \right) T_a^{2e} E_1 T_a^{-2e} \tag{5.5}$$

selects first nonzero base-four address digit one in position e. Here each base-four address digit groups two successive binary clock bits, with the first bit least significant.

Add a reset projection Z, set $S = T_f T_t$, and impose

$$Z = (I - P_f) + P_f (I - P_t) S Z S^*. \tag{5.6}$$

In exact models write $Q_j^f = T_f^j P_f T_f^{-j}$, $P_{t,j} = T_t^j P_t T_t^{-j}$ and

$$C_k = \left(\prod_{j<k} Q_j^f \right) (I - Q_k^f), \qquad D_k = \prod_{j<k} (I - P_{t,j}). \tag{5.7}$$

The product defining C_k means $(\prod_{j<k} Q_j^f)(I - Q_k^f)$, with empty product I.

5.3 Counters and transitions

Encode counter value n by $r = 2^{-n}$. For each counter add a positive contraction r_i, a zero-test projection z_i and a positive contraction s_i with

$$r_i = z_i + \tfrac{1}{2}(I - z_i) s_i. \tag{5.8}$$

All instantaneous workspace variables commute. For each control state σ, add a projection q_σ, with $q_\sigma q_\tau = 0$ for $\sigma \neq \tau$ and $\sum_\sigma q_\sigma = I$. Each program branch selector is q_σ, $q_\sigma z_i$ or $q_\sigma(I - z_i)$; these selectors partition I. The next control is the sum of selectors entering that state. The next counter is the sum, over selectors, of that selector times $r_i/2$, r_i or $2r_i$ as its instruction dictates. The factor two is used only on a nonzero decrement branch. These prescriptions define a fixed rational polynomial map Step on counters and controls.

Initialization is

$$\mathrm{Init}(q_{\mathrm{start}}) = L, \quad \mathrm{Init}(q_{\mathrm{idle}}) = I - L, \quad \mathrm{Init}(r_1) = R, \quad \mathrm{Init}(r_i) = I \ (i > 1),$$

with other controls zero. For each counter or control W, impose

$$ZW = Z\,\mathrm{Init}(W), \tag{5.9}$$

$$U^*WU = Z_+\,\mathrm{Init}(W) + (I - Z_+)\,\mathrm{Step}(W), \qquad Z_+ = U^*ZU. \tag{5.10}$$

The witnesses and zero-tests satisfy (5.8) at every time; separate update equations for them are unnecessary. Add a flag projection F and require

$$q_{\mathrm{halt}} \le F. \tag{5.11}$$

5.4 Commutations and the conditional clock

The following is the complete additional commutation list.

- L, R, F commute with each other, with A_a, and with every fuel and time clock variable, including U.
- Z and all instantaneous workspace variables commute with one another and with A_a, A_f, A_t, L, R, F.
- The address set, including V, H, commutes with the complete fuel and time sets.

Workspace is not required to commute with U or the address shifts and flips.

For the conditional clock add positive contractions B, C, a projection Q and a self-adjoint unitary Y. Impose

$$Q \le F, \qquad [B,Q] = [B,F] = [C,Q] = [C,F] = [B,C] = 0, \tag{5.12}$$

$$VBV^* = 4B - 3Q, \qquad VCV^* = 4C - 3F, \tag{5.13}$$

$$YFY = F, \qquad YQY = F - Q, \tag{5.14}$$

$$[Y, VBV^*] = 0, \qquad [Y, C] = 0. \tag{5.15}$$

Each of B, C, Q, Y commutes with A_a and every fuel and time clock variable, including U. Define

$$F_i = V^i F V^{-i}, \qquad Q_i = V^i Q V^{-i}, \qquad Y_i = V^i Y V^{-i}. \tag{5.16}$$

The intended ternary measurement is $(I - F_i, F_i - Q_i, Q_i)$. Its inactive outcome contributes no required bit; its two active outcomes are exchanged by Y_i. The coordinate C records the flags. It is needed to control commutation of their translates, which commutation with A_a alone would not imply.

This finishes the finite presentation, denoted $\mathcal{P}_K$. Every branch and every variable used above occurs in a finite list; e, k, i in later arguments index consequences, not additional defining relations.

6 Every tracial limit pays for halting

Lemma 6.1 (Forced finite initialization)**.** *In an exact model of $\mathcal{P}_K$,*

$$B_e L = B_e, \quad B_e R = 2^{-e} B_e, \qquad C_k Z = C_k D_k. \tag{6.1}$$

The zero-test equation forces $z_i v = v$ when $r_i v = v$, and $z_i v = 0$ when $r_i v = 2^{-n} v$ for an integer $n > 0$.

Proof. By Lemma 4.1, every digit is a spectral function of its clock coordinate. The required commutations imply that L, R, Z commute with the relevant digits. Expand (5.3) and (5.4) along the e successive E_{00} factors of B_e. The nonrecursive terms vanish there. At the first E_1 factor, the nonrecursive term is I. The coefficient accumulated in the second expansion is 2^{-e}. These finite expansions prove the first two identities.

In (5.6), on C_k the first k fuel bits force the recursive branch. Each supplies the next factor $I - P_{t,j}$. At the next fuel bit the constant branch is forced. Thus $C_k Z = C_k D_k$. In particular the $k = 0$ identity forces reset on $I - P_f$, so $Z = 0$ cannot be a tracial solution.

For the zero-test, z_i commutes with s_i. On its range $r_i = I$; on its kernel $0 \le r_i \le I/2$. If $r_i v = v$, the latter component of v must vanish; if $r_i v = 2^{-n} v$ with $n > 0$, the former component must vanish. The argument applies to every vector in a given subspace, including after unitary conjugation. Although $r_i = 0$ is permitted, no finite sequence of the prescribed updates starting at an ordinary encoded integer reaches it. □

Theorem 6.2 (Flag density)**.** *Every exact tracial model of $\mathcal{P}_K$ satisfies $\tau(F) \ge \theta_K$. There is a function $\omega(\delta) \to 0$ such that every weighted precision-δ model satisfies*

$$\mathrm{Tr}(\rho F) \ge \theta_K - \omega(\delta). \tag{6.2}$$

No effective modulus is asserted.

Proof. Work first in the faithful tracial representation. The clock variables satisfy Lemma 4.1. By Proposition 4.5, U increments each finite time prefix; it commutes with fuel and address variables. Fix $e \in K$ with halting time h and choose k with $2^k > h$. The operator

$$Q_0 = B_e C_k D_k (I - F)$$

is a projection. Lemma 6.1 and (5.9) set all counters and controls on its range to the initialized values. Input counter one has value $r_1 = 2^{-e}$, while other counters encode zero.

Write $W_l = U^{-l} W U^l$ and $Z_l = U^{-l} Z U^l$. On Q_0, the first k time bits are initially zero. Increment does not return them to zero before step 2^k, so

$$Z_0 Q_0 = Q_0, \qquad Z_l Q_0 = 0 \quad (1 \le l \le h). \tag{6.3}$$

We prove by induction that every counter and control at time $l \le h$ has its classical scalar right action on Q_0. The assertion holds at zero by initialization. Given these right actions, the zero-test part of Lemma 6.1, applied to the conjugated workspace, forces the correct zero-test right actions as well. Every branch selector and every update polynomial therefore has its prescribed scalar right action. If $AQ_0 = aQ_0$ and $BQ_0 = bQ_0$, then $ABQ_0 = abQ_0$; no inter-time commutation is needed.

Conjugate (5.10) by U^{-l} and multiply by Q_0 on the right. The initialized polynomials are functions of L, R and constants, so retain their scalar actions. By (6.3) the reset term vanishes and the step term yields exactly the next configuration. This proves the induction. At time h,

$$U^{-h} q_{\mathrm{halt}} U^h Q_0 = Q_0.$$

Since $q_{\mathrm{halt}} \le F$ and $[F, U] = 0$, the left side is killed by $I - F$. Thus $Q_0 = 0$.

It remains to remove the auxiliary conditions, which must not be assumed to have uniform conditional mass. The operator $B_e(I - F)$ commutes with all fuel and time flips. Conjugate $B_e(I - F) C_k D_k = 0$ by the flip of the earliest specified fuel digit. It preserves every later specified digit. Adding the two equations removes that digit. Repeat in increasing order to remove all fuel conditions, then do the same for time. The result is $B_e(I - F) = 0$. The address cylinders B_e are orthogonal and have trace $4^{-(e+1)}$ by (4.3). Taking finite sums over halting inputs and then their supremum proves $\tau(F) \ge \theta_K$.

For the uniform approximate statement, suppose instead that a sequence of models has $\delta_j \to 0$ but $\mathrm{Tr}(\rho_j F) \le \theta_K - \gamma$ for some $\gamma > 0$. Form their states on the universal C^*-algebra of the finitely many bounded-domain variables before imposing equations. A weak-* limit of moment states exists. It is tracial, since

$$\|[\rho_j, s]\|_1 \le 2\|[\sqrt{\rho_j}, s]\|_2 \longrightarrow 0$$

for each generator and hence each fixed word. Here $\|\cdot\|_1$ denotes trace norm; the inequality follows by expanding $[\xi^2, s]$ and applying Hilbert–Schmidt Cauchy–Schwarz. For every relation r, the limit has $\tau(r^*r) = 0$. Traciality gives, for bounded a,

$$\tau(a^*r^*ra) = \tau(r^*raa^*) \le \|a\|^2\tau(r^*r) = 0.$$

Thus all relations are zero operators in the tracial GNS representation. This exact model has $\tau(F) \le \theta_K - \gamma$, contradicting the first part. Taking the supremum of the positive shortfall over precision-δ models defines a function $\omega(\delta) \to 0$ and proves (6.2). □

The proof treats only finite computations before taking compactness. It never tries to quantify how far a given finite precision sees into the halting set. This compactness argument supplies no effective modulus for the uniform flag-density estimate.

7 Simultaneous finite models

Theorem 7.1. *Let $h(e)$ be the recognizer's halting time, or ∞ if it does not halt. For every even integer $N \ge 2$, define*

$$K_N = \{e < N/2 : h(e) < 2^N\}, \qquad \theta_N = \sum_{e \in K_N} 4^{-(e+1)}.$$

There is a flat model of $\mathcal{P}_K$ on dimension

$$d_N = 2^{3N+k_N}, \qquad k_N = \theta_N N + O(\log(N+2)), \tag{7.1}$$

with every defining defect at most $3 \cdot 4^{-N}$ in operator norm. The coefficients θ_N increase to θ_K.

Proof. Take N-bit address, fuel and time registers, with the finite coordinate (4.6). Their bit zero is least significant. Let the address integer be $x \in \mathbb{Z}/2^N\mathbb{Z}$ and put

$$f_N(x) = \sum_{e \in K_N} 1_{\{x \equiv 4^e \pmod{4^{e+1}}\}}. \tag{7.2}$$

The residue classes are disjoint and their periods divide 2^N, so this is a Boolean function of mean θ_N.

Every interval of N consecutive addresses has

$$\sum_{j=1}^{N} f_N(x-j) = N\theta_N + O(\log(N+2)) \tag{7.3}$$

uniformly in x, including cyclic intervals. To prove it, put $m = \lfloor \log_4 N \rfloor$. For each $e \le m$, one residue class has a count differing from its mean by at most one. Every flagged address from $e > m$ is a multiple of 4^{m+1}, so the whole remaining family contributes at most one address to such an interval. Its expected count is at most $N/(3 \cdot 4^{m+1})$. Summing proves (7.3).

Set

$$k_N = 1 + \max_x \sum_{j=1}^{N} f_N(x-j) \tag{7.4}$$

and add k_N optional qubits. Let R_0 rotate their contents from slot i to slot $i+1$ cyclically. Define

$$V|x,y\rangle = |x+1, R_0^{f_N(x)} y\rangle, \qquad F = f_N(x), \quad Q = FP_{\text{slot }0}, \quad Y = (I-F) + FX_{\text{slot }0}. \tag{7.5}$$

Define the translates F_j, Q_j by (5.16) and set

$$B = \sum_{j\geq 0} 3\cdot 4^{-j-1} Q_j, \qquad C = \sum_{j\geq 0} 3\cdot 4^{-j-1} F_j. \tag{7.6}$$

These are explicit finite diagonal matrices. Indeed V is a permutation of a finite basis, so the series are periodic geometric sums. They are positive contractions. Their terms commute, so (5.12) is exact. Shifting the sums proves (5.13) exactly; the controlled flip gives (5.14) exactly.

At an active Q_j and a fixed current address x, its occupied slot is

$$\sum_{r=x-j}^{x-1} f_N(r) \pmod{k_N}.$$

If $1 \leq j \leq N$, this sum is positive because $f_N(x-j) = 1$, and is at most $k_N - 1$ by (7.4). Thus Y preserves Q_j for those j. It always commutes with C, and the uncontrolled remainder satisfies

$$\|[Y, VBV^*]\| \leq 2 \sum_{j>N} 3\cdot 4^{-j} = 2\cdot 4^{-N}. \tag{7.7}$$

This verifies (5.15).

To couple V to the address clock, let T_0, X_0 be its ordinary cyclic rotation and first-bit flip and put

$$h_N(x) = \sum_{0\leq r<x} f_N(r), \quad D = \bigoplus_x R_0^{h_N(x)}, \quad T_a = D(T_0\otimes I)D^*, \quad X_a = D(X_0\otimes I)D^*. \tag{7.8}$$

The coordinate A_a and digit P_a commute with D, so all address clock defect bounds from Theorem 4.4 are unchanged. In the D^* gauge, V is ordinary increment except at the all-ones address, where it carries a fixed optional-register permutation. The recursive increment agrees away from that address. At all ones its two output addresses are distinct, with the same optional permutation. The guard is 4^{-N} there. Hence its operator defect is $\sqrt{2}\,4^{-N}$ and its flat defect is $\sqrt{2}\,4^{-N}2^{-N/2}$.

For $H(x)$ use 4^{-k} when the address has $k < N$ initial ones and then a zero, and use zero at all ones. Rotation removes one initial one, so (5.1) is exact, including its all-ones case. All commutations in (5.2) are exact, and off all ones

$$A_a(x+1) - A_a(x) = -1 + \tfrac{7}{4}\,4^{-k}.$$

At all ones the defect of the last equation is 4^{-N}. Its flat norm is $4^{-N}2^{-N/2}$. Let U be ordinary cyclic time increment and use ordinary clocks on fuel and time. Their defects have the same bounds as before.

It remains to verify the whole program on these same registers. For a nonzero address let e be the position of its first nonzero base-four digit. Set $R(x) = 2^{-e}$ and $L(x) = 1$ precisely when that digit is one. At the all-zero address set $R = L = 0$. Since N is even, rotation by

two bits removes one whole base-four digit. Thus (5.3)–(5.4) hold exactly, including at the all-zero address.

For fuel with k initial ones then a zero, put

$$Z(q,t) = 1_{\{t \equiv 0 \pmod{2^k}\}}.$$

On all-ones fuel use $k = N$. When the first fuel bit is zero, $Z = 1$. Otherwise the factor $I - P_t$ in (5.6) requires the first time bit to be zero and the rotated equation tests the remaining $k-1$ bits. On all-ones fuel, the recursive test is true precisely at the all-zero time word. This proves (5.6) in every case.

At a basis point (x, q, t) assign every workspace counter and control the deterministic configuration

$$\mathrm{Step}^{t \bmod 2^k}(\mathrm{Init}(x)). \tag{7.9}$$

These are diagonal entries on existing registers; they add no quantum tensor factor. For ordinary counter entries set $z_i = 1, s_i = 0$ when $r_i = 1$ and $z_i = 0, s_i = 2r_i$ otherwise; the idle $r_i = 0$ case uses $z_i = s_i = 0$. These are contractions and give (5.8) exactly. The next time point is initialized if its residue modulo 2^k is zero, and is otherwise one step of the old configuration. This is exactly (5.9)–(5.10), also at the cyclic time boundary. A displayed halt occurs in fewer than 2^N steps on an input $e < N/2$, so its address is flagged in (7.2). Hence (5.11) is exact.

All workspace and reset variables are diagonal in (x, q, t), and the optional register carries no workspace labels. Every specified remaining commutation therefore holds. The only nonzero defining defects are the clock affine and tail-flip defects, the guarded increments, the carry step and (7.7). Their operator norms are bounded by the theorem's constant. The three compulsory registers and the optional register have dimension 2^{3N+k_N}, and (7.3) gives (7.1). Finally each halting input eventually meets both cutoffs in K_N, while the address weights have a summable geometric tail. This proves $\theta_N \uparrow \theta_K$. □

8 Counting measurements on the actual state

The count must tolerate measurements that commute only approximately and a density matrix that is not flat. Its proof keeps the measurement branches together. Its last step is the usual random-access-code argument [30], using Holevo's information bound [17].

Theorem 8.1 (Triangular measurement count)**.** *Let m PVMs on $\mathbb{C}^d$ have two or three outcomes. Let ρ be a density matrix and $\xi = \sqrt{\rho}$. Assume each outcome has state commutator at most η, and outcomes of different measurements have weighted commutator at most η. For each measurement i, assume a unitary U_i has state commutator at most η, exchanges its own two binary outcomes, or its two active ternary outcomes while fixing the inactive one, and preserves every later measurement. Each of these intertwining equations is required in weighted norm with error at most η.*

If b measurements are binary and F_i is the active projection of a ternary measurement, then for an absolute constant C, with $e = Cm^4\eta^2 \le 1/2$,

$$\log d \ge b + \sum_{i \ \mathrm{ternary}} \mathrm{Tr}(\rho F_i) - Cm^3\eta - m\big[h_2(e) + e\big]. \tag{8.1}$$

Here h_2 is binary entropy, with $h_2(0) = 0$.

Proof. For a PVM (P_a) define left and right Hilbert–Schmidt isometries

$$L_P(Z) = (P_a Z)_a, \qquad R_P(Z) = (Z P_a)_a.$$

They differ on ξ by at most $\sqrt{3}\eta$. Every left multiplication commutes with every right multiplication. Moving a product of s left recording maps through ξ to right maps in reverse order therefore changes the recording vector by at most $s\sqrt{3}\eta$, by telescoping isometries.

Suppose a column of left operators $D = (D_c)_c$ has column operator norm at most L and $\|D\xi\|_2 \le \alpha$. It commutes with all right recording maps. After any prefix of s measurements, the preceding comparison gives

$$\left(\sum_{\mathbf{a},c} \|D_c P_{s,a_s} \cdots P_{1,a_1}\xi\|_2^2\right)^{1/2} \le \alpha + Ls\sqrt{3}\eta. \tag{8.2}$$

This estimates all branches together. In particular, no inverse of a small branch probability appears.

Measure in reverse index order. For the suffix starting at i, move U_i through its recorded measurements. It exchanges its own two labels and preserves later ones. A single intertwining column is a difference of two isometries, has column operator norm at most two and weighted norm $O(\eta)$ on ξ. Equation (8.2) bounds it after any preceding measurements by $O(m\eta)$. At most m intertwiners, followed by the initial state-invariance error, give total vector error $O(m^2\eta)$. Distances of purifications bound total variation of the classical records. Thus the suffix distribution differs from its own i-th outcome swap by at most $Cm^2\eta$.

For a binary outcome with conditional law $(p, 1-p)$, concavity gives $h_2(p) \ge 1 - |2p-1|$. The suffix swap variation is exactly the average of $|2p-1|$ over the later record. Hence its conditional entropy is at least one minus that variation. For ternary probabilities,

$$H(p_0, p_1, p_2) \ge (p_1 + p_2) - |p_1 - p_2|.$$

Indeed split off the active event and apply the binary inequality within it. The ternary swap variation is the corresponding average of $|p_1 - p_2|$. The chain rule now bounds the full record entropy below by b plus recorded active probabilities minus $Cm^3\eta$.

Those recorded active probabilities can be replaced by their values on ρ. For one PVM, $(P_a\xi)_a$ and $(\xi P_a)_a$ purify, respectively, its nonselectively measured state and ρ on the original system. Their distance is at most $\sqrt{3}\eta$, so those states differ by trace norm at most $C\eta$. If Φ_j are the nonselective measurement channels, expand

$$\Phi_s \cdots \Phi_1(\rho) - \rho = \sum_{j=1}^{s} \Phi_s \cdots \Phi_{j+1}(\Phi_j(\rho) - \rho).$$

Trace-norm contractivity gives an $O(m\eta)$ bound for every prefix. Thus all active expectations change by at most $Cm^2\eta$ in total, already covered by the preceding error term.

We next bound record entropy above by dimension. Fix one recorded variable. Move its measurement to the end of the actual sequence by adjacent swaps. An elementary swap column has components $[P_a, Q_b]$, weighted norm $O(\eta)$ and column operator norm at most two: it is the difference of the two recording isometries, with record factors reordered. Equation (8.2) bounds its norm after a prefix by $O(m\eta)$. After at most m swaps the total recording vectors differ by $O(m^2\eta)$.

In the reordered process this PVM is last. Remeasuring it on the final system recovers its recorded value with zero error, since equal projective measurements are adjacent. In the original process, apply the same failure projection to the difference of recording vectors. Its squared norm bounds the decoding error by $e = Cm^4\eta^2$.

Let Z be the classical record and Q_{fin} the final system, of dimension d. The decoder just described acts only on this system, not on the purification reference. Holevo's bound,

conditional entropy subadditivity and the error-indicator bound for at most three outcomes give

$$\begin{aligned} H(Z) &= I(Z : Q_{\mathrm{fin}}) + H(Z \mid Q_{\mathrm{fin}}) \\ &\le \log d + \sum_i H(Z_i \mid Q_{\mathrm{fin}}) \\ &\le \log d + m[h_2(e) + e]. \end{aligned}$$

For the last line, measuring the proposed decoder can only increase conditional entropy; the classical conditional entropy is at most $h_2(e) + e \log(3-1)$. Combining the upper and lower record bounds proves (8.1). □

9 The matched lower bound

Theorem 9.1. *Every weighted precision-δ model of $\mathcal{P}_K$ on $\mathbb{C}^d$ satisfies*

$$\log d \ge (3 + \theta_K) \log_4(1/\delta) - o(\log(1/\delta)). \tag{9.1}$$

The error is uniform over all states and dimensions.

9.1 The carry estimate

Identify $\mathcal{K}$ with binary sequences viewed as 2-adic integers, and let g_i add the nonnegative integer i. If two sequences first differ at digit j, their Cantor coordinates differ by between $\frac{1}{2}4^{-j}$ and 4^{-j}. Addition preserves that first differing digit. Consequently every g_i is 2-Lipschitz on $\mathcal{K}$, uniformly in i.

Let $h(a) = 4^{-k}$ if the sequence starts with exactly k ones followed by a zero, and put $h(1) = 0$. It is also 2-Lipschitz: equal first-zero positions give equal values; otherwise the smaller first-zero position j is the first differing digit, and the value difference is at most 4^{-j}. Its identities are

$$h = 1 - p + \tfrac{1}{4} p(h \circ s), \qquad g_1(a) = a - 1 + \tfrac{7}{4} h(a). \tag{9.2}$$

The latter follows by summing the coordinate change when k initial ones become zero and the next zero becomes one. The all-ones case follows by the limit $k \to \infty$. Lipschitz functions on $\mathcal{K}$ can be extended linearly across its gaps with the same constant.

Lemma 9.2 (Translations with polynomial cost)**.** *For an absolute constant C, every model of the address relations satisfies*

$$|H - h(\widehat{A}_a)|_\rho \le C\delta, \tag{9.3}$$

$$|V^{-i} \widehat{A}_a V^i - g_i(\widehat{A}_a)|_\rho \le C(i+1)\delta, \tag{9.4}$$

$$|[V^i J V^{-i}, P_{a,j}]|_\rho \le C(i+1)4^j \delta \quad (J \in \{F, Q, Y\}). \tag{9.5}$$

Proof. Let $p = p(\widehat{A}_a)$ and $E = H - h(\widehat{A}_a)$. Subtract the first identity of (9.2) from (5.1), retaining its error r. The exact subtraction is

$$\begin{aligned} E = & -(P_a - p) + \tfrac{1}{4} P_a T_a E T_a^* \\ & + \tfrac{1}{4} P_a (T_a h(\widehat{A}_a) T_a^* - h(s(\widehat{A}_a))) \\ & + \tfrac{1}{4} (P_a - p) h(s(\widehat{A}_a)) + r. \end{aligned}$$

The first, third, fourth and last terms have weighted size $O(\delta)$ by Lemmas 2.3 and 4.2. In the fourth term, moving the bounded right factor past ξ costs $O(\delta)$, since $h \circ s$ is Lipschitz with

fixed constant. The second term is bounded by $\frac{1}{4}|E|_\rho + O(\delta)$ using $\|E\| \le 1$. Absorb it to prove (9.3). The last equation of (5.2), together with spectral repair and unitary transport, gives

$$|V^* \widehat{A}_a V - g_1(\widehat{A}_a)|_\rho \le C_1 \delta.$$

Put $e_i = |V^{-i} \widehat{A}_a V^i - g_i(\widehat{A}_a)|_\rho$, so $e_0 = 0$. Since $g_i = g_{i-1} \circ g_1$ on $\mathcal{K}$ and functional calculus commutes with unitary conjugation, for $i \ge 1$

$$\begin{aligned} V^{-i} \widehat{A}_a V^i - g_i(\widehat{A}_a) = V^* (V^{-(i-1)} \widehat{A}_a V^{i-1} - g_{i-1}(\widehat{A}_a)) V \\ + (g_{i-1}(V^* \widehat{A}_a V) - g_{i-1}(g_1(\widehat{A}_a))). \end{aligned}$$

The bracket in the first term is a difference of positive contractions, so it has norm at most one, and Lemma 2.2 bounds the first term by $e_{i-1} + \|[V, \xi]\|_2 \le e_{i-1} + \delta$. Both $V^* \widehat{A}_a V$ and $g_1(\widehat{A}_a)$ have spectrum in $\mathcal{K}$, where g_{i-1} is 2-Lipschitz. Hence (2.3), the bound $\|[g_1(\widehat{A}_a), \xi]\|_2 \le 2\|[\widehat{A}_a, \xi]\|_2$ and (4.4) bound the second term by $2(C_1 + 4C)\delta$. Thus $e_i \le e_{i-1} + C_2\delta$ with C_2 independent of i, and induction gives $e_i \le C_2 i\delta$. This proves (9.4). Each step pays one transport of the accumulated error and one functional-calculus loss with the uniform constant 2; no iterated factor four occurs.

Put $f_j = p \circ s^j$. Replace $P_{a,j}$ by $f_j(\widehat{A}_a)$, at cost $C4^j\delta$ by (4.5), and conjugate by V^{-i}, at transport cost $i\delta$. Then apply (2.3) to (9.4), with the Lipschitz constant $2 \cdot 4^j$ of f_j. The result is

$$|V^{-i} P_{a,j} V^i - f_j(g_i(\widehat{A}_a))|_\rho \le C(i+1)4^j\delta.$$

The composition $f_j \circ g_i$ has Lipschitz constant at most $4 \cdot 4^j$ on $\mathcal{K}$. Each $J \in \{F, Q, Y\}$ commutes with A_a to precision δ, and hence with $\widehat{A}_a$ to precision $C\delta$. Equation (2.4) bounds its commutator with this composition by $C4^j\delta$. Replace back at cost $C(i+1)4^j\delta$: for the self-adjoint difference Δ just estimated, $|[J, \Delta]|_\rho \le 2|\Delta|_\rho + \|\Delta\|\|[J, \xi]\|_2$. Conjugate by V^i, at transport cost $2i\delta$. This gives (9.5). □

9.2 Joint digits and conditional measurements

Lemma 9.3. *Fix $M \ge 1$. The $3M$ compulsory binary measurements are the raw digit PVMs $(P_{c,j}, I - P_{c,j})$, where $P_{c,j} = T_c^j P_c T_c^{-j}$ for $c \in \{a, f, t\}$ and $j < M$, with flips $X_{c,j} = T_c^j X_c T_c^{-j}$. The M conditional ternary measurements are $(I - F_i, F_i - Q_i', Q_i')$ for $i < M$, with flips Y_i, where Q_i' is the repaired data projection constructed in the proof. Index the conditional measurements first, then the address, fuel and time digits, each family in increasing digit order. All commutations between outcomes of different measurements, all flip equations required in this order, and all state commutations have error at most*

$$\eta = C(M+1)^6 4^M \delta. \tag{9.6}$$

Each Q_i' differs from Q_i by weighted size at most $C\eta$, and the conditional measurements are exact PVMs on the same state.

Proof. Two elementary rules are used throughout. First, if D is self-adjoint and Y is a contraction, then $DY\xi = D\xi Y + D[Y, \xi]$ gives

$$|[D, Y]|_\rho \le 2|D|_\rho + \|D\| \, \|[Y, \xi]\|_2. \tag{9.7}$$

Hence, if $X = X_0 + D_X$ and $Y = Y_0 + D_Y$ with self-adjoint corrections, the identity

$$[X, Y] = [D_X, Y] + [X_0, D_Y] + [X_0, Y_0]$$

shows that replacing both sides of a commutator by nearby operators adds the two replacement errors; it never multiplies them. Second, suppose every letter of a word w commutes to

precision δ with every letter of a word w', adjoint letters included. Moving letters one at a time and applying Lemma 2.2 to each interchange gives $|[w, w']|_\rho \le C|w|\,|w'|(|w| + |w'| + 1)\delta$. We call this the word-interchange bound.

The required estimates are the following.

(i) *One compulsory clock.* Lemma 4.2 gives the digit commutations, own-bit flips, later-digit preservation and state commutators within each clock, with error $C(M+1)^3 4^M \delta$. Its raw digit commutators already replace both digits by commuting spectral digits, at additive cost.

(ii) *Different compulsory clocks.* For $c \neq c'$, $P_{c,j}$ and $P_{c',k}$ are words of lengths $2j+1$ and $2k+1$ in two complete clock sets, which commute by hypothesis. The word-interchange bound gives $|[P_{c,j}, P_{c',k}]|_\rho \le C(M+1)^3\delta$. The same bound shows that $X_{c,j}$ preserves every $P_{c',k}$. These estimates must go through the raw conjugated digits. Comparing the spectral digits $p(s^j(\widehat{A}_c))$ and $p(s^k(\widehat{A}_{c'}))$ by two applications of (2.4) would multiply their Lipschitz constants. That costs order $4^{j+k}\delta$, which for j, k near M exceeds (9.6) by a factor of order $4^M/(M+1)^6$ and would defeat the choice (9.8) below. If spectral digits were measured instead, put $D_{c,j} = p(s^j(\widehat{A}_c)) - P_{c,j}$. It is self-adjoint, with norm at most one and $|D_{c,j}|_\rho \le C4^j\delta$ by (4.5). Then

$$[p(s^j(\widehat{A}_c)), p(s^k(\widehat{A}_{c'}))] = [D_{c,j}, p(s^k(\widehat{A}_{c'}))] + [P_{c,j}, D_{c',k}] + [P_{c,j}, P_{c',k}].$$

By (9.7), with $\|[p(s^k(\widehat{A}_{c'})), \xi]\|_2 \le C4^k\delta$ from Lemma 2.3 and $\|[P_{c,j}, \xi]\|_2 \le (2j+1)\delta$, its weighted norm is at most $C4^M\delta + C(M+1)^3\delta$. Thus either choice satisfies (9.6); we fix the raw digits.

(iii) *Conditional against conditional.* Use the affine-only part of the spectral repair on (B, Q, V) and (C, F, V). If $i \le j$,

$$[Q_i, F_j] = V^i [Q, F_{j-i}] V^{-i}.$$

Recover F_{j-i} from the rounded flag coordinate and apply (2.4) with $[Q, C] = 0$. The error is $O(4^{j-i}\delta)$ plus polynomial transport. If $i > j$, use $[F, B] = 0$ instead. The same argument with $[Q, B] = 0$ controls $[Q_i, Q_j]$, and with $[F, C] = 0$ controls $[F_i, F_j]$. Only one side is replaced by a coordinate function here, so there is a single depth loss.

(iv) *Conditional against address digits.* By (9.5) with $J \in \{F, Q\}$, the commutators of F_i and Q_i with $P_{a,j}$ are at most $C(M+1)4^M\delta$. With $J = Y$, the same bound shows that Y_i preserves every address digit.

(v) *Conditional against fuel and time digits.* The operators F_i, Q_i and Y_i are words of length $2i+1$ in V, V^* and one of F, Q, Y. Each of these letters commutes with every fuel and time clock variable by hypothesis. The digits $P_{f,k}$ and $P_{t,k}$ are words of length $2k+1$ in the fuel and time sets. The word-interchange bound gives commutators at most $C(M+1)^3\delta$, and shows that Y_i preserves every fuel and time digit. These bounds are again taken through the raw conjugated words. Writing F_i or Q_i as a digit of a conditional coordinate, and a fuel or time digit as a spectral digit, would stack two depth losses.

(vi) *Flips.* The own conditional flip is the V^i-translate of (5.14). For later flags, $[Y, C] = 0$ and digit recovery imply preservation by Y_i. For later data, use $[Y, VBV^*] = 0$ and digit recovery on the tail coordinate, with depth $j - i - 1$ when $j > i$. This again costs at most one factor 4^{j-i}. Conditional flips preserve all compulsory digits by (iv) and (v). A compulsory flip $X_{c,j}$ exchanges $P_{c,j}$ and preserves later digits of its own clock by (i), and every digit of a later clock family by (ii).

(vii) *State commutators.* State commutators of translated raw variables are at most $C(M+1)\delta$.

All the estimates just listed fit (9.6): cubic word-interchange costs and linear address costs are dominated by its sixth power. No factor 4^{i+j} is used.

We finally repair inclusion. For projections F, Q with $|(I-F)Q|_\rho \le \eta$ and state commutators at most η, put

$$A_0 = FQF, \quad L_0 = (I-F)QF, \quad Q' = 1_{[1/2,1]}(A_0) \le F.$$

The weighted error rule and adjoints give $|Q - A_0|_\rho \le 4\eta$ and $|L_0|_\rho \le 2\eta$. Since $A_0 - A_0^2 = L_0^* L_0$ and

$$|1_{[1/2,1]}(t) - t|^2 \le t(1-t) \quad (0 \le t \le 1),$$

we get $|Q' - A_0|_\rho \le 2\eta$ and thus $|Q' - Q|_\rho \le 6\eta$. This difference is self-adjoint, so its state commutator is at most twice its weighted norm. Replacing Q by Q' in a commutator or a flip equation therefore costs only $C\eta$, by Lemma 2.2. Apply this independently to the raw pairs F_i, Q_i, after absorbing their polynomial transport constants into (9.6). The outcomes $(I - F_i, F_i - Q_i', Q_i')$ are now exact PVMs. All previous estimates survive with a fixed factor. □

Proof of Theorem 9.1. Index the optional measurements first, then the three compulsory clock families, in increasing digit order within each family. Optional flips preserve all compulsory measurements. A compulsory flip needs to preserve only later compulsory measurements, not the earlier optional family. Thus this ordering meets the hypotheses of Theorem 8.1, with $m = 4M$ and $b = 3M$.

Put $t = \log_4(1/\delta)$ and, for sufficiently small δ, choose

$$M = \lfloor t - 30 \log_4(2+t) \rfloor . \tag{9.8}$$

Then $4^M \delta \le (2+t)^{-30}$, so (9.6) gives $\eta \le C(1+t)^{-24}$. The error terms of (8.1) are $o(1)$. Consequently

$$\log d \ge 3M + \sum_{i<M} \operatorname{Tr}(\rho F_i) - o(1).$$

Unitary state transport gives

$$|\operatorname{Tr}(\rho F_i) - \operatorname{Tr}(\rho F)| \le \|V^{-i} \rho V^i - \rho\|_1 \le 2i\delta.$$

Apply Theorem 6.2 to the same ρ. The result is

$$\log d \ge (3 + \theta_K - \omega(\delta))M - o(M).$$

Since $M = t - O(\log(t+2))$ and $\omega(\delta) \to 0$, this is (9.1). Every estimate used the actual model state. □

Remark 9.4 (Where the guards are used). Two structural facts about the proof follow from tracing where each relation enters.

(a) The guarded relations (4.7) are used only in the exact tracial limit. In the proof of Theorem 1.2 they enter only through Proposition 4.5, in the exact part of Theorem 6.2, and the approximate part of that theorem reaches the exact case by compactness, not by a quantitative estimate. The quantitative lower bound of this section uses instead the unguarded carry equation, the last relation of (5.2), whose defect in the finite models is 4^{-N} (Theorem 7.1). Consequently Proposition 4.6 is not used in the proof of Theorem 1.2, and its guard-removal loss, of order $\delta^{1/6}$ up to logarithms, is never paid.

(b) The guarded increment (4.7) for the address increment V is used by no proof. Section 6 applies Proposition 4.5 only to U, and V acts on the address digits only through the carry relations (5.1)–(5.2), in Lemma 9.2. That relation is therefore redundant. It is harmless: the finite models satisfy it with operator defect $\sqrt{2}\,4^{-N}$ (Theorem 7.1).

10 The game and its exponent

Corollary 10.1 (Matched source profiles)**.** *The presentation $\mathcal{P}_K$ satisfies*

$$\log W(4^{-N}) = (3+\theta_K)N + o(N) = \log H(4^{-N}). \tag{10.1}$$

One computable comparison sequence is $A_N = 3N + \lfloor N\theta_N \rfloor$, and both logarithmic profiles equal $A_N + o(N)$.

Proof. Theorem 9.1 is the lower bound for W, and $W \le H$. Theorem 7.1 is the matching upper bound for H at a fixed multiple of the requested precision. Replacing N by an even integer at a fixed greater distance absorbs that multiple. Such a change costs $O(1)$ depth, and the upper and lower limiting slopes are both $3+\theta_K$ since $\theta_N \to \theta_K$. Monotonicity interpolates intermediate depths. Finally $N\theta_N = N\theta_K + o(N)$, giving the comparison sequence. □

The Boolean correspondence theorem [12, Theorem 4.9] supplies both game directions with the actual marginal state kept. We do not reprove that general correspondence or its return theorem. To make the precision accounting for this construction transparent, we give the elementary matrix estimate responsible for its square-root extraction and explain how it applies here.

Lemma 10.2 (Consistency and the square root of the state)**.** *Let A, B be reflections and Ξ a rectangular amplitude matrix with $\rho_B = \Xi\Xi^*$. If $\|B\Xi - \Xi A\|_2 \le \eta$, then*

$$\|[B, \sqrt{\rho_B}]\|_2 \le \sqrt{2}\,\eta. \tag{10.2}$$

Proof. Set $Y = (\Xi + B\Xi A)/2$. Then $BY = YA$, so B commutes with YY^* and its square root. Also $\|\Xi - Y\|_2 \le \eta/2$. For any rectangular matrices M, N,

$$\|\sqrt{MM^*} - \sqrt{NN^*}\|_2 \le \sqrt{2}\,\|M - N\|_2. \tag{10.3}$$

Indeed apply the eigenbasis proof of (2.2) to $f(t) = |t|$ and the self-adjoint block matrices $\left(\begin{smallmatrix} 0 & M \\ M^* & 0 \end{smallmatrix}\right)$ and its N analogue, with $Z = I$. Their absolute values are block diagonal with square-root blocks. The right-hand squared norm is $2\|M-N\|_2^2$. Discard the second left-hand block to obtain (10.3). Since B commutes with $\sqrt{YY^*}$, its commutator with $\sqrt{\rho_B}$ is bounded by twice their difference, proving (10.2). □

Proposition 10.3 (The two transfers needed here)**.** *Let G be the fixed context-variable game supplied by Theorem 3.1 for $\mathcal{P}_K$. There are constants $c, C > 0$ such that a loss-ε strategy with local dimensions at most d gives a weighted source model of precision $C\sqrt{\varepsilon}$ in dimension at most cd. A flat source model of precision δ in dimension d gives a strategy of dimension at most cd and loss at most $C\delta^2$.*

Proof. The upper direction is precisely the explicit flat strategy in Theorem 3.1. For the lower direction apply the established Boolean correspondence, followed by that theorem's weighted soundness. The correspondence is applicable because every compiled context is an ordinary Boolean predicate with a nonempty set of valid answers and every tested incidence has fixed positive probability.

Here is also a direct check of the precision ingredients in its extraction, without invoking its more general return theorem. A mixed strategy may first be replaced by a pure component of at least the same winning probability. Compressing to Schmidt supports gives local dimension at most d. Alice's context POVMs have simultaneous fixed-size ancilla dilations, and Bob's binary contractions have reflection dilations on a common extra qubit. The dimensions grow by a fixed factor; the initial ancilla state is included in the marginal. Invalid Alice answers can be reassigned to a fixed valid answer, since an invalid answer was already a certain loss. After these reductions, consistency at a tested incidence gives

$$\|(A \otimes I - I \otimes B)\psi\| \le 2\sqrt{\varepsilon/\pi_{\min}}.$$

Pull a fixed Bob context word across to Alice in reverse order. Every replacement costs this bound, and Alice's context observables commute and satisfy their truth table exactly. Thus each Bob context commutator and forbidden assignment polynomial has error $C\sqrt{\varepsilon}$ on the marginal. Lemma 10.2 gives the same order for its state commutators, with no additional square root. These are exactly the Boolean-model conditions used by the compiler. Hence the composed source precision is $C\sqrt{\varepsilon}$ on that state. □

Proof of Theorem 1.2. Apply Proposition 10.3 and Corollary 10.1. For $\varepsilon = 2^{-n}$, precision $C\sqrt{\varepsilon}$ has

$$\log_4(1/(C\sqrt{\varepsilon})) = n/4 + O(1).$$

The weighted lower profile therefore gives

$$\log D_G(2^{-n}) \ge \tfrac{3+\theta_K}{4} n - o(n).$$

The flat models at precision 4^{-N} give loss at most $C16^{-N}$ and logarithmic dimension $(3+\theta_K)N + o(N)$. Choosing an even $N = n/4 + O(1)$ at or above the required precision gives the matching upper bound. Fixed dimension multipliers add only $O(1)$ to the logarithm. Monotonicity also gives (1.4) for arbitrary $\varepsilon \downarrow 0$. These strategies have losses tending to zero, so $\omega_q(G) = 1$. A finite-dimensional perfect strategy would extract a source model at every positive precision in one fixed dimension, contradicting Theorem 9.1.

An oracle for K computes θ_K by finite summation and a geometric tail bound. Conversely, suppose membership below e is known and subtract those contributions. With $w_e = 4^{-(e+1)}$, the residual is at most $w_e/3$ when $e \notin K$, and at least w_e when $e \in K$. An approximation of error less than $w_e/6$, compared with threshold $2w_e/3$, distinguishes these cases. Induction computes K from a real oracle for θ_K. Thus $\theta_K \equiv_T K$, and the same holds for $\beta = (3+\theta_K)/4$. Lemma 2.4 now excludes a computable certified convergence rate. □

11 Every positive left-c.e. exponent

The halting game uses separated address cylinders so that its coefficient has an explicit Turing degree. To realize an arbitrary left-c.e. real, the program must instead read the whole address. The clocks, reset, carry and conditional measurement estimates stay the same.

11.1 A finite reader

Fix a computable increasing sequence of rationals $0 \le q_s \le 1$ tending to $\theta \in [0,1]$. For a binary address $x = (b_0, b_1, \ldots)$ put

$$v(x) = \sum_{j \ge 0} b_j 2^{-j-1}, \qquad p_s = \sum_{j<s} b_j 2^{s-1-j}.$$

At stage $s \geq 1$, a fixed program computes q_s, reads the next bit, updates p_s, and halts if

$$(p_s + 1)/2^s < q_s. \tag{11.1}$$

Counter instructions implement the integer arithmetic and strict rational comparison. Each stage takes at least one instruction. Acceptance implies $v(x) \leq (p_s + 1)/2^s < \theta$; conversely, if $v(x) < \theta$, the two sequences $(p_s+1)/2^s \to v(x)$ and $q_s \to \theta$ eventually satisfy (11.1). Thus the program halts exactly on $\{v < \theta\}$. Only the finite program for (q_s) occurs in the presentation; θ is not an operator coefficient.

Here is the full change to Section 5. Delete L, R, equations (5.3)–(5.4), and their occurrences in initialization and commutation lists. Retain all other clock, reset, carry and conditional relations. Add positive contractions $A_{\mathrm{r}}, C_{\mathrm{r}}$ and a projection P_{r} to the instantaneous commuting workspace, with

$$4A_{\mathrm{r}} = 3P_{\mathrm{r}} + C_{\mathrm{r}}, \qquad [P_{\mathrm{r}}, C_{\mathrm{r}}] = 0. \tag{11.2}$$

The workspace commutes with Z, A_a, A_f, A_t, F as before. The flag still commutes with A_a and the complete fuel and time sets, including U; all the conditional-clock commutations are retained. Initialize every ordinary counter to I, the start control to I, the other controls to zero, and A_{r} to A_a. A READ control q_σ has selectors $q_\sigma P_{\mathrm{r}}$ and $q_\sigma(I - P_{\mathrm{r}})$. If Q_{r} is the sum of READ controls, set

$$\mathrm{Step}(A_{\mathrm{r}}) = Q_{\mathrm{r}} C_{\mathrm{r}} + (I - Q_{\mathrm{r}}) A_{\mathrm{r}}. \tag{11.3}$$

Include A_{r}, together with every ordinary counter and control, in (5.9)–(5.10). In particular,

$$ZA_{\mathrm{r}} = ZA_a, \qquad U^* A_{\mathrm{r}} U = Z_+ A_a + (I - Z_+)\,\mathrm{Step}(A_{\mathrm{r}}).$$

The bit and tail are witnesses, like the zero-test witnesses; they need no separate update equations. Call this finite rational presentation $\mathcal{P}_\theta$.

Lemma 11.1 (Reader forcing)**.** *Every exact tracial model of $\mathcal{P}_\theta$ has $\tau(F) \geq \theta$. Uniformly over its weighted precision-δ models,*

$$\mathrm{Tr}(\rho F) \geq \theta - \omega_\theta(\delta), \qquad \omega_\theta(\delta) \longrightarrow 0. \tag{11.4}$$

Proof. On the kernel and range of P_{r}, equation (11.2) places the spectrum of A_{r} in $[0, 1/4]$ and $[3/4, 1]$, respectively. Therefore

$$P_{\mathrm{r}} = p(A_{\mathrm{r}}), \qquad C_{\mathrm{r}} = s(A_{\mathrm{r}}).$$

This conclusion does not require Cantor spectrum in advance.

Let B_w specify an s-bit address word w satisfying (11.1). Every extension follows the same computation to a halt, at some time h. It may halt at an earlier stage; in either case it halts by stage s after at most s reads. Choose $2^k > h$ and put $Q_0 = B_w C_k D_k (I - F)$. These factors commute. The reset part of Lemma 6.1 is independent of the deleted decoder and initializes this subspace.

The reader requires an operator induction in addition to the scalar counter induction of Theorem 6.2. After j reads, track

$$(A_{\mathrm{r}})_\ell Q_0 = s^j(A_a) Q_0. \tag{11.5}$$

The right side commutes with Q_0. Taking adjoints shows that Q_0 reduces the self-adjoint left side. Functional calculus on that reducing subspace gives

$$(P_{\mathrm{r}})_\ell Q_0 = b_j Q_0, \qquad (C_{\mathrm{r}})_\ell Q_0 = s^{j+1}(A_a) Q_0$$

whenever the next bit is read. All bits read before the halt belong to w. On other instructions the reader remains fixed. The ordinary counters and controls have their prescribed scalar right actions, with zero tests forced by Lemma 6.1. There is no reset before h. Thus (5.10) proves the initialized history through its halt. As in Theorem 6.2, $q_{\mathrm{halt}} \le F$ and $[F, U] = 0$ force $Q_0 = 0$. Conjugating by successive fuel and time flips and adding removes C_k, D_k, giving $B_w \le F$.

The length-s address cylinders have trace 2^{-s}. The number accepted at stage s is

$$a_s = \max\{0, \lceil 2^s q_s \rceil - 1\}, \qquad 2^{-s} a_s \ge q_s - 2^{-s}.$$

Summing accepted cylinders gives $\tau(F) \ge q_s - 2^{-s}$, hence $\tau(F) \ge \theta$. This includes $\theta = 0$ and $\theta = 1$: the possibly unaccepted all-ones address has trace zero. Finally the bounded moment-limit and tracial GNS argument in Theorem 6.2 applies to all the added bounded variables and proves (11.4). □

Lemma 11.2 (Finite reader models)**.** *For even $N \ge 2$, $\mathcal{P}_\theta$ has flat models of dimension*

$$d_N = 2^{3N + k_N}, \qquad k_N = \theta N + o(N), \tag{11.6}$$

with every defining residual at most $3 \cdot 4^{-N}$ in operator norm. Every new program and reader equation is exact.

Proof. Use the three N-bit registers of Theorem 7.1, with bit zero least significant for increment. Write

$$v_N(x) = \sum_{j<N} b_j(x) 2^{-j-1}, \quad \vartheta_N = q_{2^N}, \quad f_N(x) = 1_{\{v_N(x) < \vartheta_N\}}.$$

Read the N bits followed by zeros. The Cantor reader and its witnesses remain exact after any number of reads. Assign all workspace entries the history (7.9), with this initialization. The next history point is reset precisely when its time residue is zero, including the cyclic boundary and all-ones fuel. Thus all transitions are exact. A displayed halt has accepting stage $s \le 2^N$, so $v_N(x) < q_s \le q_{2^N}$; hence $q_{\mathrm{halt}} \le F$ is exact. Specifying q_{2^N} may take arbitrarily long classical computation. It does not lengthen a displayed history or add a tensor factor.

The moving-window estimate holds for this new mask. On an aligned block of 2^r consecutive address integers, the radical-inverse values v_N are $(j + u)/2^r$, $0 \le j < 2^r$, for a fixed $u \in [0, 1)$. Their count below ϑ_N differs from $2^r \vartheta_N$ by at most one. Decompose an ordinary interval of length L into at most two aligned dyadic blocks of each size, and a cyclic interval into at most two ordinary intervals. Uniformly in the starting point,

$$\sum_{j=1}^{N} f_N(x - j) = N \vartheta_N + O(\log(N + 2)). \tag{11.7}$$

This is the elementary radical-inverse discrepancy argument associated with van der Corput [39]. It is proved here for the moving windows used by the construction.

Use this f_N and $k_N = 1 + \max_x \sum_{j=1}^{N} f_N(x - j)$ in (7.5)–(7.8). The slot noncollision, geometric series and dressing arguments in Theorem 7.1 use only a Boolean mask and this slot bound. Their residual estimates therefore still hold. Since $\vartheta_N \to \theta$, equation (11.7) gives (11.6). The full address mean differs from ϑ_N by at most 2^{-N}, so these flat models also have $\mathrm{Tr}(\rho F) \to \theta$. □

11.2 Adding clocks and powering equations

Proposition 11.3. *For every $r \in \mathbb{N}$, integer $q \geq 1$ and left-c.e. $\theta \in [0,1]$, there is a fixed finite rational BCS game $G_{\theta,r,q}$ of quantum value one, with no finite-dimensional perfect strategy, and*

$$\log D_{G_{\theta,r,q}}(\varepsilon) = \frac{3+r+\theta}{4q} \log(1/\varepsilon) + o(\log(1/\varepsilon)). \tag{11.8}$$

Proof. Append r copies of the compulsory clock, requiring each complete new generator set to commute with all generators outside that clock. Tensoring the upper models adds rN to logarithmic dimension. For the lower bound, append the new compulsory digit families to the order in Lemma 9.3. Its word-interchange argument bounds all new cross terms by a polynomial in M times δ. The same error form $C_r(M+1)^6 4^M \delta$ holds, with $m = (4+r)M$ and $b = (3+r)M$. Neither the carry argument nor any other quantitative step uses the removed decoder. Applying Theorem 8.1 and Lemma 11.1, at depth (9.8), gives

$$\log W(\delta) \geq (3+r+\theta) \log_4(1/\delta) - o(\log(1/\delta)). \tag{11.9}$$

Indeed the intermediate count is $(3+r)M + \sum_{i<M} \operatorname{Tr}(\rho F_i) - o(1)$, and state transport bounds the sum below by $M(\theta - \omega_\theta(\delta)) - O(M^2\delta)$.

Replace every defining polynomial equation $p = 0$, including specified cross-commutators, by $h_q(p) = 0$, where

$$h_q(p) = \begin{cases} (p^*p)^{q/2}, & q \text{ even}, \\ p(p^*p)^{(q-1)/2}, & q \text{ odd}. \end{cases} \tag{11.10}$$

Keep the exact variable domains and the weighted state-commutator conditions. Since $h_q(p)^* h_q(p) = (p^*p)^q$, scalar Jensen applied to the probability measure $B \mapsto \operatorname{Tr}(\rho 1_B(p^*p))$ gives

$$\|p\sqrt{\rho}\|_2 \leq \|h_q(p)\sqrt{\rho}\|_2^{1/q}. \tag{11.11}$$

No commutation between ρ and p^*p is needed. A powered precision-η model is consequently an original precision-$\eta^{1/q}$ model, also for its state commutators when $\eta \leq 1$. Conversely the operator-norm bounds of Lemma 11.2 give

$$\|h_q(p)\| = \|p\|^q \leq 3^q 4^{-qN},$$

and the flat state commutators vanish. All powered equations are fixed rational $*$-polynomials. Theorem 3.1 and the game transfer give original precision $(C\sqrt{\varepsilon})^{1/q}$ for the lower bound, and upper loss $C16^{-qN}$. Taking an even $N = \log(1/\varepsilon)/(4q) + O(1)$ above the required precision proves (11.8). All compiler dimension multipliers are constant for each fixed presentation. The upper losses tend to zero; the lower bound diverges, excluding a finite-dimensional perfect strategy. □

Dimension part of Theorem 1.1. Given a positive left-c.e. α, choose integers $q \geq 1$ and $m \geq 3$ with $m \leq 4q\alpha < m+1$. Put $r = m-3$ and $\theta = 4q\alpha - m \in [0,1)$. This is left-c.e.; an increasing rational name is $\max(0, 4q\alpha_s - m)$. Proposition 11.3 gives exponent α. The integers q, m are chosen once for the given α. The usable-entanglement assertion is proved in Section 12. □

Corollary 11.4. *Uncomputable dimension exponents of fixed finite BCS games accumulate at zero. In fact powering the original halting presentation gives exponents $(3+\theta_K)/(4q)$, each Turing-equivalent to K.*

Proof. Apply (11.11) and the operator upper estimate of Theorem 7.1 to $\mathcal{P}_K$. The invertible rational affine map preserves Turing degree, and these positive exponents tend to zero. □

12 Usable entanglement of the original state

Dimension alone does not give a distillation protocol. The separable state $d^{-1}\sum_x |xx\rangle\langle xx|$ has marginal I/d and no distillable entanglement. The measurement count of Section 8 must therefore be supplemented by agreement between the players' actual operators. This section extends the binary triangular extraction of [12, Theorem 6.2] to measurements with an inactive outcome. Keeping the random targets preserves the mean number of active Bell pairs.

12.1 Triangular extraction with an inactive outcome

For operators D_ℓ and a vector ψ, write

$$\mathcal{R}_\psi((D_\ell)_\ell) = \left(\sum_\ell \|D_\ell\psi\|^2\right)^{1/2}.$$

Opposite-party tensor factors are suppressed below.

Lemma 12.1 (Extraction on a mixed state)**.** *Let ψ_{ABE} purify a finite-dimensional state ρ_{AB}. For $1 \le j \le m$, each party $D = A, B$ has an exact three-outcome PVM $(P^D_{j,\perp}, P^D_{j,0}, P^D_{j,1})$. Bob has a reflection U^B_j and Alice a unitary U^A_j. Put $F^D_j = P^D_{j,0} + P^D_{j,1}$. Suppose each of the following quantities is at most η:*

(i) For every j, $\mathcal{R}_\psi((P^A_{j,a} - P^B_{j,a})_a)$ and $\|(U^A_j - U^B_j)\psi\|$.

(ii) For each $s < j$, with $a, b \in \{\perp, 0, 1\}$, the two separate quantities $\mathcal{R}_\psi(([P^B_{j,a}, P^B_{s,b}])_{a,b})$ and $\mathcal{R}_\psi(([U^B_j, P^B_{s,b}])_b)$.

(iii) For every j, with $b \in \{0, 1\}$, $\mathcal{R}_\psi((U^B_j P^B_{j,b} U^B_j - P^B_{j,1-b})_b)$.

Set $\kappa = 7m^3\eta$ and $g(t) = (1+t)h_2(t/(1+t))$. If $\kappa \le 1/2$, then

$$E_D^{A\to B}(\rho_{AB}) \ge \sum_{j=1}^m \mathrm{Tr}(\rho_B F^B_j) - \tfrac{1}{2}m(m-1)\eta - 6m\kappa - 2g(\kappa). \tag{12.1}$$

A compulsory binary stage is included by $P^D_{j,\perp} = 0$. Different controls need not commute.

Proof. One stage. Alice samples a uniform bit a and sends it to Bob. Each party applies the local isometry

$$W^a_D = |0,0\rangle P^D_\perp + \sum_{b=0,1} |1,b\rangle (U_D)^{a\oplus b} P^D_b. \tag{12.2}$$

The first recorded bit is activity and the second is a coherent output qubit. Orthogonality of the three labels gives $(W^a_D)^* W^a_D = I$. A purification of the actual random operation is

$$\mathcal{A}\zeta = 2^{-1/2}\sum_a |a,a,a\rangle (W^a_A \otimes W^a_B)\zeta.$$

The third target copy records classical randomness in a new environment. It does not act on E. Compare this with the isometry

$$\mathcal{I}\zeta = \sum_a |a,a,a\rangle \left[2^{-1/2}|00\rangle_{\text{flags}}|00\rangle_{\text{qubits}} P^B_\perp \zeta + |11\rangle_{\text{flags}}|\Phi_2\rangle P^B_a \zeta\right]. \tag{12.3}$$

Its squared branch norms sum to $\|\zeta\|^2$.

For an arbitrary input ζ, let p, u, c be its projection-mirror aggregate, control-mirror norm and active-swap aggregate. Then

$$\|(\mathcal{A} - \mathcal{I})\zeta\| \le p + (u + c)/\sqrt{2}. \tag{12.4}$$

To check this, put $V_{a,\perp} = I$, $V_{a,b} = U^{a\oplus b}$ and use the exact identity

$$V^A_{a,r}P^A_r = P^B_r V^B_{a,r} + V^A_{a,r}(P^A_r - P^B_r) + P^B_r(V^A_{a,r} - V^B_{a,r}).$$

Multiplication by Bob's $V^B_{a,o}P^B_o$ kills the first term unless $o = r$. Equal inactive outcomes give $P^B_\perp$; equal active outcomes give P^B_a, apart from one swap defect when $r = 1 - a$. Summing squares over all records bounds the three error families by p, $u/\sqrt{2}$, and $c/\sqrt{2}$. For example $\sum_o \|V^B_{a,o}P^B_o z\|^2 = \|z\|^2$, and the control error survives only at $r = 1 - a$. Minkowski proves (12.4). No swap condition on Alice is used.

All histories. The ideal residual Kraus operators on either party are

$$R_{j,a,0} = P_{j,\perp}/\sqrt{2}, \qquad R_{j,a,1} = P_{j,a}.$$

They are self-adjoint and $\sum_{a,f} R^2_{j,a,f} = I$. For a history of j stages let K^B_h be the chronological product on Bob and Q^A_h the reversed product on Alice. Summing the innermost outcome repeatedly, without interchanging measurements, gives

$$\sum_h K^{B*}_h K^B_h = \sum_h K^B_h K^{B*}_h = I, \tag{12.5}$$

and both analogous identities for Q^A_h. The recursion

$$R^B K^B_h - Q^A_h R^A = R^B(K^B_h - Q^A_h) + Q^A_h(R^B - R^A)$$

therefore implies

$$\Delta_j := \mathcal{R}_\psi((K^B_h - Q^A_h)_{|h|=j}) \le j\eta.$$

For any family of Bob operators D_ℓ, opposite-party commutation and completeness give the useful estimate

$$\left(\sum_{h,\ell} \|D_\ell K^B_h \psi\|^2\right)^{1/2} \le \mathcal{R}_\psi((D_\ell)_\ell) + \left(\sum_\ell \|D_\ell\|^2\right)^{1/2} \Delta_j. \tag{12.6}$$

Indeed write $D_\ell K^B_h \psi = Q^A_h D_\ell \psi + D_\ell(K^B_h - Q^A_h)\psi$ and sum squares.

Expand a current operator's commutator with a history one factor at a time. Later outcomes disappear by completeness. At a prefix of length s, equation (12.6) bounds each commutator column by $(1 + Ls)\eta$. For the three current projections one can take $L = 3/2$, since commutators of two projections have norm at most $1/2$. For the current control take $L = \sqrt{3}$, since a unitary's commutator with a projection has norm at most one. The two scaled inactive copies count as one in the squared sum. Thus, at a stage preceded by j stages $(0 \le j < m)$, their history commutator aggregates are at most

$$(j + \tfrac{3}{4}j(j-1))\eta \le 3j^2\eta, \qquad (j + \tfrac{\sqrt{3}}{2}j(j-1))\eta \le 2j^2\eta.$$

For a mirror use $(G_A - G_B)K^B_h = K^B_h(G_A - G_B) - [G_B, K^B_h]$, rather than inserting a bipartite defect into (12.6). Its two aggregates obey

$$p_j \le (3j^2 + 1)\eta, \qquad u_j \le (2j^2 + 1)\eta.$$

The two Bob swap defects have operator root norm at most $2\sqrt{2}$, so (12.6) gives $c_j \le (1+2\sqrt{2}j)\eta$.

The ideal prefix is an orthogonal sum of histories, with normalized Bell or inactive output factors. Apply (12.4) to these histories together. The error of the stage preceded by j stages is at most

$$[(3+\sqrt{2})j^2+2j+1+\sqrt{2}]\eta \le 7(j+1)^2\eta.$$

Telescoping the compositions, with actual isometries on the left and ideal prefixes on the right, bounds the final vector difference by $\sum_{j=0}^{m-1} 7(j+1)^2\eta \le 7m^3\eta = \kappa$. Completeness, rather than the number of histories, controls every sum.

Mean activation. Discarding the ideal stage's records gives Bob the channel $\Lambda_j(X) = \sum_r P^B_{j,r} X P^B_{j,r}$. The two recording vectors $(P^B_{j,r}\psi)_r$ and $(P^A_{j,r}\psi)_r$ are unit vectors at distance at most η. Their Bob marginals are $\Lambda_j(\rho_B)$ and ρ_B, respectively. Hence $\|\Lambda_j(\rho_B)-\rho_B\|_1 \le 2\eta$. Contractivity and telescoping bound a prefix disturbance by $2(j-1)\eta$. For a traceless self-adjoint X and $0 \le F \le I$, $|\operatorname{Tr}(FX)| \le \|X\|_1/2$. The ideal number N_{act} of active outputs therefore has

$$\mathbb{E}N_{\mathrm{act}} \ge \sum_j \operatorname{Tr}(\rho_B F^B_j) - \tfrac{1}{2}m(m-1)\eta. \tag{12.7}$$

The targets remain random throughout this argument. Selecting one fixed target could select a different activation density.

Hashing the actual output. Discard the original systems and targets and locally dephase the activity flags. Keep the output qubits coherent. The ideal output is

$$\sigma = \sum_f p_f |f,f\rangle\langle f,f| \otimes \bigotimes_{j=1}^{m} \begin{cases} \Phi_2, & f_j = 1, \\ |00\rangle\langle 00|, & f_j = 0. \end{cases}$$

Its coherent information is $I(A\rangle B)_\sigma = \mathbb{E}N_{\mathrm{act}}$: $S(B) = H(p) + \mathbb{E}N_{\mathrm{act}}$ and $S(AB) = H(p)$. The actual output ω has half trace distance at most κ from σ. Each local output space has dimension 4^m.

For states on dimension D at half trace distance t,

$$|S(\alpha) - S(\beta)| \le t \log D + g(t). \tag{12.8}$$

For a direct proof write $\alpha-\beta = \Delta_+ - \Delta_-$ with $\operatorname{Tr}\Delta_\pm = t$ and compare the two binary-mixture decompositions of $(\alpha+\Delta_-)/(1+t) = (\beta+\Delta_+)/(1+t)$. Concavity and the binary-mixture upper bound give (12.8) in both directions. Also g is increasing. Apply the bound to B and AB, of dimensions 4^m and 16^m. Then

$$I(A\rangle B)_\omega \ge \mathbb{E}N_{\mathrm{act}} - 6m\kappa - 2g(\kappa).$$

The hashing inequality [9, Theorem 10] makes this an achievable Alice-to-Bob rate. All preprocessing was one-way LOCC on ρ_{AB}, so composing it with hashing lower-bounds $E_D^{A\to B}(\rho_{AB})$. Together with (12.7) this proves the lemma. □

12.2 The bridge from an actual mixed strategy

Proposition 12.2. *For a loss-ε strategy of the compiled game of any presentation containing the clock, carry and conditional-clock relations of Section 5 and r extra compulsory clocks, powered by q (in particular G_K, the games of Proposition 11.3 and those of Theorem 15.7), the hypotheses of Lemma 12.1 hold on its original mixed state, after adjoining pure local ancillas, with $m = (4+r)M$ and*

$$\eta = C_{\mathcal{P},r,q}(M+1)^6 4^M \delta, \qquad \delta = (C_{\mathcal{P},r,q}\sqrt{\varepsilon})^{1/q}. \tag{12.9}$$

The active expectations are $(3+r)M + \sum_{i<M} \operatorname{Tr}(\rho_B F_i)$ *for the reconstructed source model on the actual Bob marginal.*

Proof. The actual-state interface is [12, Proposition 6.1]; the relevant precision ingredients are also visible in the compiler and Lemma 10.2. Here they must be applied before any pure-component selection. Relabel illegal Alice answers to a valid answer and dilate the finitely many POVMs with pure local ancillas. On a purification ψ_{ABE} of the original state, fix for each variable one incident Alice context. Its reflection A_v and Bob's reflection B_v satisfy

$$\|(A_v - B_v)\psi\| \le 2\sqrt{\varepsilon/\pi_{\min}} =: \mu.$$

This statement is valid with the inaccessible E retained. Moving a Bob word to Alice reverses its order and costs at most its length times μ. For a fixed polynomial certificate, transfer its right factors, then its context relation, which vanishes exactly on Alice. This gives all Boolean residuals $O(\mu)$. Lemma 10.2, with the rectangular amplitude matrix for AE against B, gives the square-root-state commutators at the same scale. Theorem 3.1(a) reconstructs the powered source on this actual marginal. Equation (11.11) gives the unpowered precision δ in (12.9). Pure local ancillas do not change distillable entanglement.

The raw measurements and controls in Lemma 9.3 are word polynomials of length $O(M)$ in Bob's reflections. Alice uses their opposite word polynomials. These are exact projections and reflections for the raw translated observables: for example, the opposite of $T_B^j P_B T_B^{-j}$ is $T_A^{-j} P_A T_A^j$, with T_A the opposite word for T_B. Their mirrors cost $O(M\mu) \le O(M\delta)$. Applying Lemma 10.2 with parties exchanged gives their Alice square-root-state commutators as well.

Repair each raw pair F_i, Q_i locally on both sides by $Q_i' = 1_{[1/2,1]}(F_i Q_i F_i)$. Bob's error is $O(\eta)$ by Lemma 9.3. On Alice, transferring $(I - F_i)Q_i$ reverses its order, giving the adjoint of her inclusion defect. For any operator R,

$$\|R^*\sqrt{\rho}\|_2 \le \|R\sqrt{\rho}\|_2 + \|[R, \sqrt{\rho}]\|_2.$$

The raw mirrors and state commutators therefore give Alice's inclusion defect $O(\eta)$. The repair proof in Lemma 9.3, using $FQF - (FQF)^2 = ((I-F)QF)^*((I-F)QF)$, gives $\|(Q_i' - Q_i)\sqrt{\rho}\|_2 = O(\eta)$ on each party. The triangle inequality now yields $\|(Q_i'^A - Q_i'^B)\psi\| = O(\eta)$. Thus the ternary PVMs have actual mirrors at the same scale.

Use the reverse of the count order: the last compulsory family first, each digit family reversed, then the optional indices reversed. A flip's preservation of later count measurements is exactly preservation of earlier extraction measurements. Bob's reflection converts the conjugation-form estimates into commutators at cost $O(\eta)$ by right multiplication and its state commutator. Lemma 9.3, with the extra clocks as in Proposition 11.3, supplies every swap and triangular relation. A fixed enlargement of $C_{\mathcal{P},r,q}$ changes individual errors into the finite root aggregates. Only Q_i was repaired, so the activation flags are the original F_i. This proves every assertion. □

Theorem 12.3. *The games* $G_{\theta,r,q}$ *satisfy*

$$d^{\rightarrow}_{G_{\theta,r,q}}(\varepsilon) = \frac{3+r+\theta}{4q}\log(1/\varepsilon) + o(\log(1/\varepsilon)). \tag{12.10}$$

The original halting game satisfies the same formula with coefficient $c_K = (3+\theta_K)/4$.

Proof. Put $t = \log_4(1/\delta)$ and choose M by (9.8). Then $\eta = O(t^{-24})$, $\kappa = O(t^{-21})$, $m\kappa = O(t^{-20})$, and every extraction error in (12.1) tends to zero. On every original strategy state,

$$E_D^{A\to B}(\rho_{AB}) \ge (3+r)M + \sum_{i<M} \operatorname{Tr}(\rho_B F_i) - o(1).$$

Lemma 11.1 and $|\operatorname{Tr}(\rho_B F_i) - \operatorname{Tr}(\rho_B F)| \le 2i\delta$ bound this below by $(3 + r + \theta - \omega_\theta(\delta))M - O(M^2\delta) - o(1)$. Since

$$M = \frac{1}{4q}\log(1/\varepsilon) - O(\log\log(1/\varepsilon)),$$

taking the infimum proves the lower bound. For the upper bound, $E_D^{A\to B}(\rho) \le \log d_B$ for any state on Bob dimension d_B, so $d_G^{\to}(\varepsilon) \le \log D_G(\varepsilon)$. Proposition 11.3 matches the lower coefficient. The explicit upper strategies are in fact maximally entangled, with only constant compiler overhead. For G_K, use Theorems 6.2 and 7.1 in the same argument; its retained raw measurements are identical. This also proves the left-c.e. usable-entanglement assertion of Theorem 1.1. □

These are equalities for the infimum demand. A particular successful strategy may carry arbitrary unused extra entanglement. The non-effective remainder comes from flag-density convergence; the extraction error above tends to zero absolutely.

Corollary 12.4. *Every positive left-c.e. real belongs to both coefficient sets $\mathcal{A}_{\log}$ and $\mathcal{A}_{\dim}$ of [12, Section 11.6].*

Proof. In that notation $\mathcal{A}_{\log}$ consists of the finite limits $d_G^{\to}(\varepsilon)/\log_2(1/\varepsilon)$ for finite value-one games, and $\mathcal{A}_{\dim}$ uses $\log_2 D_G(\varepsilon)$ instead. Theorem 1.1 uses these exact conventions. In particular it proves the positive-computable-real inclusion conjectured in the discussion of [12, Problem 18.3(i)], and strengthens it to positive left-c.e. reals. It is consistent with that paper's Theorem 11.14, which bounds $\mathcal{A}_{\log}$ by the reals left-c.e. relative to $0''$. □

13 Heat depends on the order of reset and communication

Use the thermodynamic model of *The heat price of precision* [10]: degenerate energy endpoints, asymptotic reversible preparation and erasure, and all working systems included in the reset. Work and bath heat coincide in its dimensionless units, with one unit equal to $k_B T \ln 2$. Write $K_{\text{ans}} = |\mathcal{A}||\mathcal{B}|$ for the product of the two answer-alphabet sizes; it is unrelated to the halting set K.

The local schedule allows no communication until both working devices are blank. The joint schedule permits quantum reunion after answer release and before reset. Theorem 2.1 of [10] gives, for every attainable loss,

$$2d_G^{\to}(\varepsilon) \le C_G(\varepsilon) \le w_{\text{local},G}(\varepsilon) \le C_G(\varepsilon) + 2\log K_{\text{ans}} \le 2\log D_G(\varepsilon) + 2\log K_{\text{ans}}, \tag{13.1}$$

where C_G is the infimum mutual information of a successful state. The same theorem gives $0 \le w_{\text{joint},G}(\varepsilon) \le 2\log K_{\text{ans}}$. These are cited results, not new erasure claims. The matching-sandwich argument of its Corollary 2.2 applies here with $o(\log(1/\varepsilon))$ remainders rather than its sharper hypothesis for the earlier coefficient family.

Corollary 13.1 (Order of operations)**.** *For every $\alpha > 0$ left-c.e., the same game G_α of Theorem 1.1 satisfies, with $L = \log_2(1/\varepsilon)$,*

$$w_{\text{local},G_\alpha}(\varepsilon) = 2\alpha L + o(L), \qquad 0 \le w_{\text{joint},G_\alpha}(\varepsilon) \le 2\log_2 K_{\text{ans}}. \tag{13.2}$$

For the original halting game the local coefficient is $(3+\theta_K)/2$, Turing-equivalent to K, while joint-reset heat stays bounded.

Proof. Substitute $d_{G_\alpha}^{\to}(\varepsilon) = \alpha L + o(L)$ and $\log D_{G_\alpha}(\varepsilon) = \alpha L + o(L)$ into (13.1). The fixed answer term is $O(1)$. The joint bound is the cited theorem. The halting coefficient is $2c_K$, an invertible rational affine image of θ_K. □

Thus the same fixed finite game and accuracy have a local-reset heat law with an uncomputable logarithmic coefficient and a bounded joint-reset cost. The two schedules differ in whether quantum reunion is allowed before reset.

14 The exact-trace barrier and matrix approximation

Let $\mathcal{T}(\mathcal{P})$ be the exact tracial states of a finite bounded rational presentation, and let $\mathcal{T}_{\mathrm{mat}}(\mathcal{P})$ be those obtained as moment limits of flat finite-dimensional models with defects tending to zero. Both spaces are viewed in the compact space of bounded word moments. A diagonal choice of approximants shows that $\mathcal{T}_{\mathrm{mat}}$ is closed. In this section the relevant space is assumed nonempty and F is a fixed rational self-adjoint polynomial. These results explain why the construction of Sections 11–12 reaches only left-c.e. targets; they do not classify all games.

14.1 The exact-trace barrier

Proposition 14.1. *Suppose every exact trace has $\tau(F) \geq \theta$, and some sequence of flat approximate models with defects tending to zero has F expectations tending to θ. Then*

$$\theta = \min_{\tau \in \mathcal{T}(\mathcal{P})} \tau(F) \tag{14.1}$$

is left-c.e. The matched fixed-observable mechanism used here realizes exactly the left-c.e. targets in $[0, 1]$.

Proof. A moment limit of the flat models is tracial and has $\tau(p^*p) = 0$ for every defining relation p. The GNS argument in Theorem 6.2 makes it an exact model. Its F expectation is θ, proving (14.1).

Here is an effective lower hierarchy for that minimum. At degree $2j$ use complex word moments, impose normalization, adjoints, cyclic invariance, a positive moment matrix, the domain localizing constraints, and every relation insertion visible at that degree. Add $|y_w| \leq 1$ for contraction words. These bounded constraints express all declared domains. Alternatively use (3.1) for positive contractions and $P = (I+x)/2$, $x = x^* = x^{-1}$, for projections. The exact encodings are finite and rational and preserve tracial models and the substituted observable's trace. The finite feasible sets are compact rational semialgebraic sets in real and imaginary coordinates. Their lower minima ℓ_j increase. A diagonal limit of minimizers is a positive tracial functional with all relations and bounded domains, so its GNS tuple is an exact model. Consequently $\ell_j \uparrow \theta$.

Real quantifier elimination decides whether $\ell_j > r$ for a rational r. Enumerate these certified lower rationals and take running maxima, starting with a rational norm lower bound for F. They converge to θ. This proves left semicomputability without an assumption about numerical SDP duality. This is the bounded tracial moment method [2, 22]; it optimizes over all exact traces. It does not identify matrix-approximable traces, just as general NPA convergence concerns commuting models [29].

Lemma 11.1 and Lemma 11.2 give the converse realization for every left-c.e. $\theta \in [0, 1]$. Fixed positive rational affine changes, extra clocks and residual powering preserve this class. □

Complementing F changes a minimum to one minus a maximum. It therefore does not reverse this one-sided forcing argument. If all exact traces gave one common value, the lower hierarchy for F and for $-F$ would make that value computable. Similarly, an irreversible acceptance program presented with a nonmonotone approximation keeps earlier accepted cylinders: their union is effectively open and its measure remains left-c.e.

14.2 Weighted limits are matrix-approximable

The next compression uses Connes' spectral-cut inequality [7], in the form Vidick uses for almost synchronous correlations [40]. Its role here is to identify possible limiting traces. Its dimension cost is never charged to the actual-state measurement bound.

Lemma 14.2 (Moment compression)**.** *Let contraction generators $X_i \in M_d(\mathbb{C})$ lie in the declared domains of Definition 2.1, and suppose $\max_i \|[X_i, \sqrt{\rho}]\|_2 \le \eta$. For each integer $l \ge 1$ and $\zeta > 0$ there is a finite-dimensional tuple Y in the same domains such that every $*$-word w of length at most l satisfies*

$$|\mathrm{tr}_D(w(Y)) - \mathrm{Tr}(\rho w(X))| \le 2l\sqrt{2\eta} + \zeta. \tag{14.2}$$

No bound on D is asserted.

Proof. Put $P_t = 1_{(t,\infty)}(\rho)$, $0 \le t \le 1$. Spectral integration gives $\rho = \int_0^1 P_t\,dt$ and $\int_0^1 \mathrm{rank}\,P_t\,dt = 1$. In an eigenbasis of ρ,

$$\begin{aligned}\int_0^1 \|[X_i, P_t]\|_2^2\,dt &= \sum_{a,b} |(X_i)_{ab}|^2 |\lambda_a - \lambda_b| \\ &\le \|[X_i, \sqrt{\rho}]\|_2 \left(2\,\mathrm{Tr}(\rho X_i X_i^*) + 2\,\mathrm{Tr}(\rho X_i^* X_i)\right)^{1/2} \le 2\eta.\end{aligned}$$

The middle inequality factors $|\lambda_a - \lambda_b|$ into the difference and sum of their square roots and applies Cauchy–Schwarz.

On $\mathrm{ran}\,P_t$ compress to $B_i = P_t X_i P_t$. Insert P_t between consecutive letters of a word of length k. Each of the $k-1$ trace errors is at most $\sqrt{\mathrm{rank}\,P_t}\,\|[X_i, P_t]\|_2$. Integrating and applying Cauchy–Schwarz bounds the total by $(k-1)\sqrt{2\eta}$.

Restore the domains inside each compressed space. Positive contractions need no repair. Round a projection at $1/2$, a reflection by spectral sign, and a unitary by polar decomposition, completing the partial isometry on its kernel. Each resulting Z_i satisfies

$$\|Z_i - B_i\|_2^2 \le \|[X_i, P_t]\|_2^2.$$

For projections use $\min(s^2, (1-s)^2) \le s(1-s)$ and $\mathrm{Tr}(B_i - B_i^2) = \|(I - P_t)X_i P_t\|_2^2$. For unitaries use $(1-s)^2 \le 1 - s^2$ and $\mathrm{Tr}(I - B_i^* B_i) = \|(I - P_t)X_i P_t\|_2^2$; spectral sign is the self-adjoint case. Telescoping the k repaired letters adds at most $k\sqrt{2\eta}$ after integration.

There are finitely many spectral cuts. The integrated repaired moments form a convex combination of normalized traces of the compressed tuples, with weights $a_h = (\mathrm{rank}\,P_h)\,\Delta t_h$ summing to one. If their dimensions are r_h, choose integer multiplicities n_h with $n_h r_h / \sum_i n_i r_i$ approximating a_h in ℓ^1 within ζ. The direct sum realizes those weights and changes every contraction-word moment by at most ζ. Combining both errors proves (14.2). □

Corollary 14.3. *Moment limits of weighted approximate models are exactly $\mathcal{T}_{\mathrm{mat}}(\mathcal{P})$. In particular, if $\lambda = \min_{\mathcal{T}_{\mathrm{mat}}} \tau(F)$, then uniformly over weighted precision-δ models,*

$$\mathrm{Tr}(\rho F) \ge \lambda - \omega(\delta), \qquad \omega(\delta) \to 0. \tag{14.3}$$

Proof. In a convergent weighted sequence pass to a subsequence with commutator errors $\eta_j \le j^{-4}$. Apply Lemma 14.2 to length j with $\zeta_j \to 0$. The resulting flat tuples have the same limiting moments. Each defining p^*p moment tends to zero, so their flat relation defects tend to zero. The reverse inclusion takes $\rho = I/d$. Compactness and the closedness of $\mathcal{T}_{\mathrm{mat}}$ give attainment of λ. A sequence violating (14.3) would have a weighted moment limit below λ, a contradiction. □

14.3 Arithmetic and the necessary trace gap

Proposition 14.4. *Set*

$$m_n = \inf\{\operatorname{tr}_d F(X) : d \geq 1,\ X \text{ a flat precision-}2^{-n} \text{ model}\}.$$

Then $\lambda = \sup_n m_n$ *is left-c.e. relative to* $0'$. *If* $a = \min_{\mathcal{T}} \tau(F)$ *and* λ *is not left-c.e., then* $a < \lambda$.

Proof. The feasible sets shrink, so m_n increases. Approximants to any trace in $\mathcal{T}_{\mathrm{mat}}$ show $m_n \leq \lambda$. Conversely choose models with costs at most $m_n + 1/n$. A moment subsequence gives a trace in $\mathcal{T}_{\mathrm{mat}}$ of cost $\lim_n m_n$, proving equality.

For fixed n, d, all domains, squared flat residuals and the cost are rational semialgebraic data; real quantifier elimination decides whether there is a feasible tuple of cost below a rational r. Enumerating d makes m_n uniformly right-c.e. More explicitly,

$$\begin{aligned} s < \lambda \iff \exists n\ \exists r \in \mathbb{Q} \quad & s < r \text{ and} \\ & \text{every dimension lacks a precision-}2^{-n} \text{ model of cost } < r. \end{aligned} \tag{14.4}$$

The last finite-dimensional predicate is decidable. The displayed lower cut is therefore Σ^0_2, or c.e. in $0'$. Proposition 14.1's hierarchy proves that a is left-c.e., and $a \leq \lambda$. A non-left-c.e. λ must consequently satisfy $a < \lambda$.

For any rational r between them there is an exact trace of cost below r, but some fixed positive precision admits no flat finite-dimensional model of cost at most r. Otherwise a sequence of such models would limit to an approximable trace below λ. This is the necessary trace gap. □

If all approximable traces have one common value, apply the same arithmetic argument to $-F$. Both cuts are then c.e. in $0'$, so the value is Δ^0_2. An arbitrary minimum above is only asserted to be left-c.e. in $0'$; that statement alone does not make it Δ^0_2.

Context: why a uniform lower hierarchy is different. Use the finite-dimensional *synchronous* value in Mousavi–Nezhadi–Yuen's hardness reduction [28, Lemma 6.7], which directly makes synchronous value one Π^0_2-hard. Place the game projection relations under a central projection p and set $F = I - p$. The scalar branch $p = 0$ always exists. The synchronous/tracial correspondence [31, 21] shows that synchronous value one gives approximable traces with $p = I$, hence minimum zero. Otherwise a positive-trace p corner would give finite synchronous strategies approaching one: compress to the representing projections, normalize by their positive limiting trace, and repair each finite PVM. For that last step, successively compress and round its first outcomes into the complement of earlier repaired projections, then use the remaining complement as the last outcome. All errors tend to zero for a fixed finite family. Thus a uniform ordinary lower hierarchy would enumerate the Σ^0_2-complete complement of synchronous value one. This contextual reduction produces only the individual minima zero and one. It uses the synchronous-value hardness itself, without a conversion from unrestricted strategies that would require extra assumptions on diagonal question weights.

Towards the next construction. The exact-trace barrier leaves open a different matched mechanism: a non-left-c.e. minimum over matrix-approximable traces, together with precision-4^{-N} flat models of dimension $2^{(3+\lambda)N+o(N)}$. The trace gap alone supplies no such upper models. Section 15 constructs them using finite perfect witnesses and an elementary slow-admission schedule. It thereby proves that the dimension spectrum strictly extends beyond the left-c.e. reals. The remaining classification questions are collected there.

15 Escaping the barrier: a complementary halting flag

The barrier in Section 14 concerns the minimum over *all* exact traces. We now construct a presentation whose minimum over matrix-approximable traces is

$$\lambda_A = 1 - \theta_A, \qquad \theta_A = \sum_{e \in A} 4^{-(e+1)},$$

for any c.e. set A. For $A = K$ this minimum is not left-c.e. The clocks retain their matched dimension budget. Every statement in this section is proved here, except the MIP* = RE input of Section 15.1.

15.1 Finite perfect witnesses as an external input

We use Ji–Natarajan–Vidick–Wright–Yuen's MIP* = RE theorem (arXiv version 4, accepted to Annals of Mathematics) in its finite perfect-completeness form [19, Theorem 12.6, pp. 193–194]. The theorem gives a computable map from machines to finite rational games: a halting machine yields a finite-dimensional perfect PCC strategy, and a nonhalting machine yields quantum value at most 1/2. Its proof starts with the trivial perfect PCC strategy of Lemma 12.3 and applies Claim 12.7 in finitely many downward steps, using the compression completeness of Theorem 12.1(1), p. 184. Only the finite-dimensional perfect PCC strategy of Theorem 12.6(1) is used: a state ψ on $H \otimes H$ and projective measurements, each player's measurements consistent on ψ in the sense of [19, Definition 5.10]. PCC means projective, consistent and commuting in that paper's sense; no symmetry between the players is assumed. The informal synchronous wording of Section 1.2 of [19] is not needed; mere quantum value one would not suffice.

Apply the map to the machine which halts precisely when $e \in A$, and denote the resulting game by $H_{A,e}$, with question sets $X_{\mathrm{A}}, X_{\mathrm{B}}$. Introduce projections p_{xa} ($x \in X_{\mathrm{A}}$) and q_{yb} ($y \in X_{\mathrm{B}}$), impose the PVM equations of both families, and impose $p_{xa}q_{yb} = 0$ for each tested losing pair. Untested pairs impose no zero-product condition. Call this finite rational projection presentation $\mathcal{A}_{A,e}$.

Lemma 15.1. *If $e \in A$, then $\mathcal{A}_{A,e}$ has an exact unital matrix model. If $e \notin A$, it has no sequence of flat matrix models whose defining defects tend to zero. Exact matrix certificates, without repeated indices, can be enumerated effectively, and their indices are exactly A.*

Proof. Let (A, B, ψ) be a finite-dimensional perfect PCC strategy and write $\psi = \mathrm{vec}(S)$. Consistency of each player's measurements reads $A_a^x S = S(A_a^x)^{\mathsf{T}}$ and $S(B_b^y)^{\mathsf{T}} = B_b^y S$; taking adjoints shows that every A_a^x and B_b^y commutes with Alice's reduced state $\rho = SS^*$. Write $\rho = \sum_c cP_c$. Its eigenspaces reduce both projection families. For a tested losing pair, consistency of Bob's measurement and perfect play give

$$0 = \langle \psi, A_a^x \otimes B_b^y \, \psi \rangle = \mathrm{Tr}(\rho A_a^x B_b^y) = \sum_{c>0} c \, \|(P_c B_b^y P_c)(P_c A_a^x P_c)\|_2^2.$$

Every summand is nonnegative. On each nonzero positive-eigenvalue eigenspace the two compressed families are PVMs and every tested losing product vanishes, including its adjoint. Restrict to any such eigenspace to obtain the unital matrix model. The commuting clause of PCC is not needed for this argument.

Conversely, approximate projection partitions can be repaired to PVMs in the same dimension. For projections $P_1, \dots, P_k$ with $S_0 = \sum_i P_i$ and $\|S_0 - I\|_{2,d} \to 0$, where $\|X\|_{2,d}^2 = \mathrm{tr}_d(X^*X)$,

$$\sum_{i \neq j} \mathrm{tr}_d(P_i P_j) = \mathrm{tr}_d(S_0^2) - \mathrm{tr}_d(S_0) \longrightarrow 0.$$

Each term is nonnegative and equals $\|P_iP_j\|_{2,d}^2$. Keep P_1, successively compress P_i to the complement R of the previously repaired outcomes and round at $1/2$, and use the last complement as the final outcome. If $\alpha = \|(I-R)P_i\|_{2,d}$, then

$$\|P_i - RP_iR\|_{2,d} \le 2\alpha, \qquad \|1_{[1/2,1]}(RP_iR) - RP_iR\|_{2,d}^2 \le \mathrm{tr}_d(RP_iR - (RP_iR)^2) \le \alpha^2.$$

Finite induction and the approximate sum equation give corrections tending to zero. Bounded-product telescoping preserves every tested losing-product defect. Using the repaired PVMs on a maximally entangled state, with Alice using the transposed p_{xa}^{T} and Bob q_{yb}, gives losing probabilities $\mathrm{tr}_d(p_{xa}q_{yb}) = \|q_{yb}p_{xa}\|_{2,d}^2 \to 0$. This contradicts the soundness bound when $e \notin A$.

For fixed e, d, real quantifier elimination decides the rational semialgebraic matrix equations, after splitting complex entries into real and imaginary parts. A nonempty set has an effectively obtainable algebraic point. Dovetail over e, d and suppress repeated indices. The first two assertions make its output exactly A. No dimension bound computable from e is asserted. □

15.2 Packing and a fixed polynomial-testing program

For a projection presentation $\mathcal{A}$ with $m \ge 1$ generators, use two projections D, E and a unitary W. Set $G_j = W^{-j}DW^j$, $0 \le j < m$, and impose

$$\sum_{j<m} W^jEW^{-j} = I, \qquad [E, G_j] = 0. \tag{15.1}$$

Substitute EG_jE for its generators in every equation of $\mathcal{A}$, replacing the constant identity by E. Clear denominators, and call the finite integer-coefficient list consisting of (15.1) and the substituted equations $\mathcal{R}(\mathcal{A})$. In $\widehat{\mathcal{P}}_A$ these are not defining relations; the program tests them.

Lemma 15.2 (Three data generators)**.** *A d-dimensional model of $\mathcal{A}$ extends to an md-dimensional model of $\mathcal{R}(\mathcal{A})$. Conversely the E corner of every tracial model of the latter carries $\mathcal{A}$, and $\tau(E) = 1/m$.*

Proof. List the projection generators as $g_0, \dots, g_{m-1}$, including both players' families for $\mathcal{A}_{A,e}$. Take $D = \mathrm{diag}(g_0, \dots, g_{m-1})$, $E = |0\rangle\langle 0| \otimes I_d$, and let W cyclically shift the m blocks forward. Then $W^{-j}DW^j$ restricts to g_j on block zero, proving the extension, also when $m = 1$. Conversely EG_jE is a projection in the unital E corner and satisfies every substituted relation. Tracing (15.1) gives $m\tau(E) = 1$. Arbitrary models need not satisfy $W^m = I$. □

Write $m_e = \#\{p_{xa}\} + \#\{q_{yb}\}$ for the total number of projections in both families of $\mathcal{A}_{A,e}$, adding an unused projection if needed, and $\mathcal{R}_e = \mathcal{R}(\mathcal{A}_{A,e})$. The map $e \mapsto \mathcal{R}_e$ is total computable. Retain the decoder L, R, reset Z, clocks, carry and conditional-clock relations of Section 5. Replace the halting recognizer by the following fixed counter transducer, delete (5.11), and add

$$(I-L)(I-F) = 0. \tag{15.2}$$

Thus an unflagged address must be in the decoded region. Keep the scalar counter domains, zero tests, controls, initialization and transition rules; their start and input are still determined by L, R.

Add the data generators D, E, W. Require them to commute with A_a, L, R, F, Z, every fuel and time generator including U, and every instantaneous scalar workspace variable. Do not require them to commute with the address shift or flip, or with one another.

The transducer has two operator registers, a monomial M and an accumulator Σ. Represent each $S \in \{M, \Sigma\}$ by two unitary generators:

$$S = (u_S + v_S)/2. \tag{15.3}$$

Every such S is a contraction, and every finite square contraction has this form. Indeed extend its polar partial isometry to a unitary V_0, put $J = |S| + i(I - |S|^2)^{1/2}$, and take $u_S = V_0 J$, $v_S = V_0 J^*$. This model assignment does not add irrational coefficients to (15.3).

On input e, compute the finite list $\mathcal{R}_e$. For each polynomial r, express its integer coefficients as repeated signed monomials, and compute an integer $b \geq 0$ with 2^b at least their total number. Clear Σ, set $M = I$ for each monomial, build it by successive right multiplications, and add its signed value times 2^{-b} to Σ. Enter CHECK, then process the next polynomial and finally idle. The extra instruction alphabet is

$$\Sigma := 0, \quad M := I, \quad M := Md \ (d \in \{D, E, W, W^*\}), \quad \Sigma := \Sigma \pm r_b M,$$

and holding a register fixed. The ordinary counter encoding supplies $r_b = 2^{-b}$. The final idle branch clears the registers to their initialized values. Every partial monomial and accumulator is a contraction, including on flagged addresses with arbitrary data.

Counter-machine universality compiles this algorithm into one fixed finite instruction table. Its finite branch selectors q_α and updates define rational polynomials Φ_S. The polynomial lists and word lengths are program outputs, rather than additional defining equations. Put $\mathrm{Init}(M) = I$, $\mathrm{Init}(\Sigma) = 0$, and impose

$$ZS = Z\,\mathrm{Init}(S), \quad U^*SU = Z_+\,\mathrm{Init}(S) + (I - Z_+)\Phi_S, \quad Z_+ = U^*ZU, \tag{15.4}$$

$$q_{\mathrm{CHECK}}\Sigma(I - F) = 0. \tag{15.5}$$

For each of the four representing unitaries in (15.3), impose commutation with the instantaneous scalar workspace and with $A_a, A_f, A_t, L, R, F, Z$. Impose no commutation with other operator registers, data, time shifts, or fuel/time flips. Scalar workspace here excludes M, Σ. This completes a finite bounded rational presentation, denoted $\widehat{\mathcal{P}}_A$.

Lemma 15.3 (Programmed testing)**.** *Every faithful exact tracial model satisfies, for every e and $r \in \mathcal{R}_e$,*

$$B_e(I - F)r(D, E, W) = 0. \tag{15.6}$$

Proof. Fix the finite CHECK time h for this relation and choose $2^k > h$. Put $Q_0 = B_e C_k D_k (I - F)$, using (5.7). The scalar initialization lemma gives $ZQ_0 = Q_0$, with no reset before h, and forces the scalar controls and counters throughout this finite history. Non-self-adjoint registers require an extra justification: $[S, Z] = 0$ gives

$$SQ_0 = SZQ_0 = ZSQ_0 = Z\,\mathrm{Init}(S)Q_0.$$

Consequently $MQ_0 = Q_0$ and $\Sigma Q_0 = 0$ initially.

Every data word commutes with Q_0; the address digits are spectral functions of A_a in an exact clock. The data also commute with U. If $S_l = U^{-l}SU^l$ has right action $S_l Q_0 = wQ_0$, then $S_l dQ_0 = wdQ_0$ for a data letter d. A conjugated scalar selector commutes with every data word, since the data commute with that selector and with U. Its forced scalar right action therefore holds on wQ_0 as well. Induction in (15.4) proves the initialized operator history. Products do not require commuting computed registers: if $AQ_0 = aQ_0$, $BQ_0 = bQ_0$, and $[b, Q_0] = 0$, then $ABQ_0 = AQ_0 bQ_0 = abQ_0$. Before h the next-reset factor vanishes on Q_0, whose range also contains every computed right action. At CHECK,

$$\Sigma_h Q_0 = 2^{-b} r(D, E, W)Q_0, \qquad (q_{\mathrm{CHECK}})_h Q_0 = Q_0.$$

Conjugate (15.5) by U^{-h} and apply it to Q_0. The flag is time invariant, and the conjugated selector commutes with data words, so $r(D,E,W)Q_0 = 0$.

The operator $r(D,E,W)B_e(I-F)$ commutes with all fuel/time flips. Conjugate its zero product with C_kD_k by the earliest specified fuel flip and add: that flip exchanges the two outcomes of its digit and preserves later digits. Repeat through the final zero of C_k, then through D_k. This removes both cylinders and proves (15.6). No conditional trace independence is needed. Since $\mathcal{R}_e$ includes the packed relations for both families, the E corner of every nonzero unflagged cylinder therefore carries both the p_{xa} and q_{yb} PVMs and all tested losing products. □

15.3 The approximable flag minimum

Proposition 15.4. *Every $\tau \in \mathcal{T}_{\mathrm{mat}}(\widehat{\mathcal{P}}_A)$ satisfies $\tau(F) \geq \lambda_A$.*

Proof. Let $Q_e = B_e(I-F)$. It commutes with the data and, by Lemma 15.3, its corner satisfies $\mathcal{R}_e$. Tracing the first packed equation gives $\tau(Q_eE) = \tau(Q_e)/m_e$. If $\tau(Q_e) > 0$, the Q_eE corner carries $\mathcal{A}_{A,e}$, with generators given by fixed bounded polynomials $Q_eEW^{-j}DW^jE$.

This positive-trace corner inherits flat approximation. In the flat approximants, self-adjointize the polynomial for Q_eE and round at $1/2$. Its projection defect tends to zero, and for every real t,

$$|1_{[1/2,\infty)}(t) - t| \leq 2|t^2 - t|.$$

The correction tends to zero in normalized Hilbert–Schmidt norm, and the rounded projection's rank fraction tends to $\tau(Q_eE) > 0$. Compress and self-adjointize the finitely many corner generators, and round them by the same estimate. Fixed polynomial relations survive by bounded-product telescoping. Normalizing the corner trace multiplies norms by a bounded factor tending to $\tau(Q_eE)^{-1/2}$. These are flat approximate models of $\mathcal{A}_{A,e}$. Lemma 15.1 therefore gives $\tau(B_e(I-F)) = 0$ for $e \notin A$.

The decoder exhausts L in trace. Iterating (5.3) yields

$$L = \sum_{e<s} B_e + R_s, \qquad 0 \leq R_s \leq \prod_{j<s} T_a^{2j} E_{00} T_a^{-2j}, \qquad \tau(R_s) \leq 4^{-s}.$$

Indeed $L - \sum_{e<s} B_e$ is a projection whose range lies in the all-00 prefix; no commutation between rotated L and all earlier digits is assumed. Since $I - F \leq L$, the disjoint cylinders of trace $4^{-(e+1)}$ give

$$\tau(I-F) = \sum_e \tau(B_e(I-F)) \leq \sum_{e\in A} 4^{-(e+1)} = \theta_A.$$

Only the passage excluding nonmember corners uses matrix approximability. The testing identity holds for every exact trace. □

15.4 Slow admission and simultaneous finite models

Lemma 15.5 (Elementary witness scheduling)**.** *Suppose finite witnesses with positive integer costs a_j arrive in a computable stream without repeated indices. At even depth N admit its longest prefix arrived by stage N, with every index $e_j < N/2$ and total cost at most $\lfloor\sqrt{N}\rfloor$. These prefixes are nested and eventually contain every fixed witness. Their total cost is $o(N)$.*

Proof. For a fixed prefix its arrival time, largest index and total cost are finite constants. All three cutoffs eventually exceed them, and none decreases with N. The cost bound is $\sqrt{N} = o(N)$. An enormous early witness may cause an enormous finite delay; no convergence rate is claimed. More generally any nondecreasing computable unbounded budget $o(N)$ has the same property. □

Apply this lemma to the certificates $(e_j, d_j, \mathrm{model}_j)$ of Lemma 15.1, with $a_j = \lceil \log_2 \max(2, m_{e_j} d_j) \rceil$. Use a dovetailing with finitely many outputs per stage. Let S_N be the admitted indices. Then $S_N \uparrow A$, and the common data factor

$$\mathcal{H}_{\mathrm{data},N} = \bigotimes_{e_j \in S_N} (\mathbb{C}^{m_{e_j}} \otimes \mathbb{C}^{d_j}), \qquad \ell_N = \dim \mathcal{H}_{\mathrm{data},N}$$

has $\log \ell_N \le \sqrt{N}$; the empty product has dimension one.

Lemma 15.6 (Complementary finite models)**.** *For even $N \ge 2$, $\widehat{\mathcal{P}}_A$ has flat models with all defining residuals at most $3 \cdot 4^{-N}$ in operator norm and*

$$d_N = 2^{3N+k_N} \ell_N, \qquad k_N = \lambda_N N + O(\log(N+2)), \qquad \lambda_N = 1 - \sum_{e \in S_N} 4^{-(e+1)} \downarrow \lambda_A. \tag{15.7}$$

Every new defining equation is exact, and $\mathrm{tr}_{d_N}(F) = \lambda_N$.

Proof. On the N-bit address cycle set

$$f_N(x) = 1 - \sum_{e \in S_N} 1_{\{x \equiv 4^e \pmod{4^{e+1}}\}}, \qquad k_N = 1 + \max_x \sum_{j=1}^{N} f_N(x-j). \tag{15.8}$$

The classes are disjoint, and their periods divide 2^N. Bit zero is least significant for increment, so these are precisely the cylinders B_e. The mask is Boolean, has mean λ_N, and satisfies (15.2). For $s = \lfloor \log_4 N \rfloor$, each selected class with $e \le s$ has interval-count error at most one. The disjoint union of the remaining classes lies in the multiples of 4^{s+1} and has at most one point in an interval of length $N < 4^{s+1}$; its expected count is at most $N/(3 \cdot 4^{s+1})$. Hence, uniformly in the subset and starting point,

$$\left| \sum_{j=1}^{N} f_N(x-j) - N\lambda_N \right| \le s + 2 + \frac{N}{3 \cdot 4^{s+1}}. \tag{15.9}$$

Cyclic intervals lift to integer intervals because all periods divide the cycle.

Use the three compulsory N-qubit registers, the k_N optional qubits, and the common data factor. Theorem 7.1's conditional permutation, series and dressing require only a Boolean mask and that k_N exceed every N-window count. They therefore apply to (15.8). On address $x \in B_{e_j}$ for an admitted index, assign the packed witness to factor j and tensor identities on the others. On other addresses take $(D, E, W) = (0, I, I)$. These address-block data are independent of fuel, time and optional slots. They commute with A_a, L, R, F, Z and all scalar workspace, and with the complete fuel/time sets. Address dressing acts on address and optional slots and leaves these commutations intact. The data need not commute with the dressed address shift or flip.

For fuel with k initial ones, use the actual initialized scalar and operator history at time $t \bmod 2^k$; all-ones fuel uses $k = N$. On each (x, q, t) block the scalar workspace is a scalar multiple of the data identity. Every M, Σ is a contraction by the signed-term bound. Choose their two representing unitaries within each block and take direct sums. All required representing-unitary commutators hold; naming these matrices adds no tensor factor.

When the next time residue is zero, the transition prescribes initialization; otherwise it prescribes the next instruction. Since $2^k \mid 2^N$, the cyclic transition from $2^N - 1$ to zero is also a reset. Thus (15.4) holds exactly at every ordinary and reset boundary. On an unflagged block every completed CHECK tests the packed witness belonging to that address, so it is zero. An unfinished CHECK imposes no test there; on flagged blocks $I - F = 0$. Therefore (15.5) is exact regardless of program speed.

The address wraparound is handled as in Theorem 7.1. In the gauge of the dressing (7.8), V is ordinary increment except at the all-ones address, where it also applies the fixed optional permutation $R_0^{\sum_x f_N(x)}$. There its output address differs from that of the recursive increment, so the guarded defect is $\sqrt{2}\,4^{-N}$ in operator norm, as before.

All new equations are exact simultaneously. The remaining clock, guarded-increment, carry and conditional-tail residuals have their old bounds, at most $3 \cdot 4^{-N}$. Counting the registers gives (15.7), and

$$\log d_N = (3+\lambda_N)N + O(\log(N+2)) + O(\sqrt{N}) = (3+\lambda_A)N + o(N).$$

A fixed increase of even depth absorbs the constant three. Bounded gaps between even depths suffice for every small precision. Computing a certificate or its admission stage is classical model-description work; it consumes neither simulated fuel nor an extra quantum register. □

15.5 Dimension, usable entanglement and heat

Theorem 15.7. *For every c.e. set A, integer $r \geq 0$ and integer $q \geq 1$, there is a fixed finite rational BCS game $\widehat{G}_{A,r,q}$ of quantum value one and with no finite-dimensional perfect strategy. With*

$$\beta_{A,r,q} = \frac{4 + r - \theta_A}{4q}, \qquad L_\varepsilon = \log_2(1/\varepsilon),$$

its resource laws are

$$\log_2 D_{\widehat{G}_{A,r,q}}(\varepsilon) = \beta_{A,r,q} L_\varepsilon + o(L_\varepsilon), \tag{15.10}$$

$$d^{\rightarrow}_{\widehat{G}_{A,r,q}}(\varepsilon) = \beta_{A,r,q} L_\varepsilon + o(L_\varepsilon), \tag{15.11}$$

$$w_{\mathrm{local},\widehat{G}_{A,r,q}}(\varepsilon) = 2\beta_{A,r,q} L_\varepsilon + o(L_\varepsilon), \qquad 0 \leq w_{\mathrm{joint},\widehat{G}_{A,r,q}}(\varepsilon) \leq 2\log_2 K_{\mathrm{ans}}. \tag{15.12}$$

The heat statement uses exactly the model and units of Section 13.

Proof. The actual-state flag bound. Moment compactness of Lemma 15.6's flat models supplies a matrix-approximable trace of flag expectation λ_A. Proposition 15.4 gives the reverse inequality, so

$$\min_{\mathcal{T}_{\mathrm{mat}}(\widehat{\mathcal{P}}_A)} \tau(F) = \lambda_A. \tag{15.13}$$

Corollary 14.3 now proves, uniformly on every original weighted precision-δ model,

$$\mathrm{Tr}(\rho F) \geq \lambda_A - \omega_A(\delta), \qquad \omega_A(\delta) \longrightarrow 0. \tag{15.14}$$

Indeed a fixed positive shortfall along $\delta \to 0$ would have a weighted moment limit in $\mathcal{T}_{\mathrm{mat}}$ below (15.13). The bound holds on the original state ρ; compression only identifies the limit class.

Dimension. Every relation used by Lemma 9.3 remains in the presentation. For $t = \log_4(1/\delta)$ and $M = \lfloor t - 30\log_4(2+t) \rfloor$, its joint errors are at most $C(M+1)^6 4^M \delta$. Theorem 8.1 and state transport give

$$\log d \geq 3M + \sum_{i<M} \mathrm{Tr}(\rho F_i) - o(M) \geq (3 + \lambda_A - \omega_A(\delta))M - O(M^2\delta) - o(M),$$

since $|\mathrm{Tr}(\rho F_i) - \mathrm{Tr}(\rho F)| \leq 2i\delta$. This is $(3+\lambda_A)\log_4(1/\delta) - o(\log(1/\delta))$. The compiler sends game loss ε to source precision $C\sqrt{\varepsilon}$, with constant dimension overhead. Lemma 15.6 gives the

converse upper loss $C'16^{-N}$. Choose an even depth $N \geq \log_{16}(C'/\varepsilon)$ within a fixed additive constant to match the bound through all small losses. This proves (15.10) for $r = 0, q = 1$.

Append r compulsory clocks with the cross-commutations of Proposition 11.3. Their upper cost is exactly rN, and their digit families add rM to the lower count. Power *every* defining residual, including the new tests and cross-commutators, by (11.10); keep domains and state-commutator requirements. Equation (11.11) turns powered precision η into unpowered precision $\eta^{1/q}$, whereas the explicit operator-norm residuals become at most $3^q 4^{-qN}$. Thus the game lower precision is $\delta = (C\sqrt{\varepsilon})^{1/q}$, upper loss is $C'16^{-qN}$, and the coefficient is $(3 + r + \lambda_A)/(4q)$. The upper losses tend to zero and this positive lower bound diverges, excluding a finite-dimensional perfect strategy.

Usable entanglement on the original mixed state. Proposition 12.2 applies. On a purification of an arbitrary mixed loss-ε strategy, keep the original environment, relabel illegal answers, and dilate locally with pure ancillas. The finite Boolean certificates reconstruct this presentation on the actual Bob marginal with powered precision $C\sqrt{\varepsilon}$; their constants depend on the fixed transducer but not on M or ε. Jensen then gives the same unpowered δ. New data/register variables enlarge a fixed certificate list only. None occurs in the raw translated clock measurements or their controls.

Consequently the same opposite-word construction on Alice, independent local inclusion repairs on both sides, and reverse count order supply the hypotheses of Lemma 12.1, with

$$m = (4 + r)M, \qquad \eta = C_{A,r,q}(M + 1)^6 4^M \delta.$$

As in Proposition 12.2, only Q_i is repaired, so the activation flags are the raw F_i on the actual marginal.

At the chosen depth $\eta = O(t^{-24})$, and the vector, activation and entropy-continuity errors of Lemma 12.1 tend to zero. Its one-way protocol and hashing therefore give on *every original strategy state*

$$\begin{aligned} E_D^{A\to B}(\rho_{AB}) &\geq (3 + r)M + \sum_{i<M} \mathrm{Tr}(\rho_B F_i) - o(1) \\ &\geq (3 + r + \lambda_A - \omega_A(\delta))M - O(M^2\delta) - o(1). \end{aligned}$$

Here (15.14), rather than an all-exact-trace bound, supplies the activation estimate. Since $M = L_\varepsilon/(4q) - O(\log L_\varepsilon)$, the infimum gives the lower side of (15.11). The general inequality $d_G^{\rightarrow}(\varepsilon) \leq \log D_G(\varepsilon)$ and the proved dimension upper bound give the other side.

Heat. Substitute the two matching laws into (13.1). Its fixed answer term is $O(1)$, so both endpoints have coefficient $2\beta_{A,r,q}$. The joint bound is the same cited heat theorem. This proves (15.12). □

Corollary 15.8 (Beyond the left-c.e. reals)**.** *The game $\widehat{G}_K := \widehat{G}_{K,0,1}$ has dimension and usable-entanglement coefficient $(4 - \theta_K)/4$, and local-heat coefficient $(4 - \theta_K)/2$. These are noncomputable right-c.e. reals, not left-c.e., and have the Turing degree of K. Write $\mathcal{B}_{\mathrm{BCS}}$ for the set of positive dimension exponents of finite BCS games with rational question probabilities and quantum value one. In particular,*

$$\{\textit{positive left-c.e. reals}\} \subsetneq \mathcal{B}_{\mathrm{BCS}} \subseteq \{\textit{positive } \Delta_2^0 \textit{ reals}\}. \tag{15.15}$$

Both $\mathcal{A}_{\log}$ and $\mathcal{A}_{\dim}$ also strictly contain the positive left-c.e. reals. The new non-left-c.e. examples accumulate at zero by taking $q \to \infty$.

Proof. The weights $w_e = 4^{-(e+1)}$ have tail at most $w_e/3$ after position e. Given earlier membership bits, absence leaves residual at most $w_e/3$, whereas presence leaves at least

w_e. Approximating θ_K within $w_e/6$ distinguishes them at threshold $2w_e/3$. Thus $K \leq_\Sigma \theta_K$. Conversely K computes the summable series, so $\theta_K \equiv_T K$. Its complement and the displayed positive rational affine images are noncomputable right-c.e.; if also left-c.e., the two rational approximations would compute them. Theorem 15.7 supplies these coefficients, proving the complementary assertion of Theorem 1.1. Its left-c.e. assertion gives the first inclusion, and Lemma 2.4 the dimension outer class. No Δ^0_2 upper bound for the entire usable-entanglement class is inferred here. □

15.6 Why the trace gap is consistent

Proposition 15.9. *For $\widehat{\mathcal{P}}_K$ the exact and approximable minima differ strictly:*

$$\min_{\mathcal{T}} \tau(F) < \min_{\mathcal{T}_{\mathrm{mat}}} \tau(F) = 1 - \theta_K.$$

An exact minimizing trace has a positive-trace nonhalting unflagged corner carrying $\mathcal{A}_{K,e}$. Such a trace is not matrix approximable.

Proof. The bounded exact-trace hierarchy of Proposition 14.1 makes the exact minimum left-c.e.; compactness supplies a minimizer. It is at most the approximable minimum and cannot equal its non-left-c.e. value. At the minimizer, $\tau(I - F) > \theta_K$. Decoder exhaustion and the total capacity of halting cylinders force $\tau(B_e(I - F)) > 0$ for some $e \notin K$. Testing and packing give its unital tracial model. Left and right multiplication in the tracial GNS representation yield a perfect commuting-operator strategy for $H_{K,e}$, using its two projection families; this is the usual tracial representation construction [31, 21]. Soundness concerns finite-dimensional value, so there is no contradiction. □

This is the matrix-versus-commuting distinction behind the negative solution of Connes' embedding problem in [19], made quantitative in the resource coefficient. Slofstra's nonclosure and undecidability results [35] are essential context for distinguishing finite exact perfection from limiting perfection; the finite exact witnesses used above are the stronger specified input, not a consequence of nonclosure.

There is also an independent consistency argument. The index set $C_{\mathrm{tr}} = \{e : \mathcal{A}_{K,e} \text{ has a unital tracial model}\}$ is Π^0_1. Each bounded tracial moment level, including domain localizers and all visible relation insertions, is a rational semialgebraic feasibility problem decidable by real quantifier elimination. Infeasibility of some level enumerates the complement; feasibility at every level gives a tracial state by bounded-moment compactness and GNS. Since $K \subseteq C_{\mathrm{tr}}$, equality would make K co-c.e. Hence some nonhalting indices, in fact infinitely many, already admit perfect tracial strategies independently of the clock construction. This argument uses the tracial presentation, rather than assuming arbitrary perfect commuting strategies are tracial. Lin's $\mathrm{MIP}^{\mathrm{co}} = \mathrm{coRE}$ [24] supplies the commuting-complexity context; no equality of commuting and finite-dimensional values is used.

15.7 Remaining classification questions and finite checks

Is every positive right-c.e. real an exponent? The construction realizes geometric complements $1 - \theta_A$, rather than arbitrary complementary flag densities. An arbitrary effectively open address set would require an unflagged address to carry one successful finite membership certificate, with simultaneous upper models and the same logarithmic budget. Testing every candidate imposes a conjunction rather than the required disjunction; waiting forever for success imposes no completed test on a rejected address. A naive integer witness can escape

to infinity in the compact counter encoding. These are missing lemmas for this route, not general impossibility results.

Is every positive d.c.e. real, a difference of two left-c.e. reals, an exponent? Independent conditional registers would need simultaneous original-state lower bounds and matched budgets for both masks and their witnesses. Subtracting two coefficient formulas does not implement subtraction on games, and arbitrary complementary flag-density realization is still missing.

Is every positive Δ_2^0 real an exponent? This precise finite-exact-witness mechanism cannot accept an arbitrary non-c.e. index set: exact finite matrix feasibility is c.e. by quantifier elimination, so finite exact witnesses for members and no flat approximating sequence for nonmembers force the set to be c.e. General limit decisions can change indefinitely. Value-one-only witnesses may grow with precision even for a fixed accepted index, need not fit an $o(N)$ logarithmic budget at precision 4^{-N}, and do not make completed CHECKs exact. A revocable or quantitatively different witness mechanism would be needed. This limitation does not strengthen the general Δ_2^0 upper bound to a spectrum classification.

The full sets $\mathcal{A}_{\log}$ and $\mathcal{A}_{\dim}$ remain undetermined; the former has the different arithmetic upper bound in [12, Theorem 11.14]. Whether a value-one game without a finite-dimensional perfect strategy can have zero logarithmic usable-entanglement coefficient, or a diverging sublogarithmic demand, remains open. A linear-system realization of these matched laws is also open.

Appendix B describes finite checks of this algebra on known-answer toy games.

How this work was done

Roles. This paper is the work of its two authors together with current frontier AI models from more than one developer. The authors set the direction throughout: they chose the questions, fixed the scope and the standards of proof and review, decided which lines to pursue and which to stop, and take full responsibility for every claim. The models did most of the exploration, derivation, computation and drafting, in a sequence of short research sprints. The written construction was preceded by separate proofs of the finite clock, guarded increment, and obstructions to exact periodic verification. A refutation-first audit then expanded the weighted carry estimates, the joint measurement count, and the simultaneous finite-model bounds. Those written proofs are incorporated here.

Validation. The evidence for the results is of two kinds at this stage: complete written proofs and computations by the scripts in `anc/`, which the reader can check. The review served to find errors; it does not replace the proofs.

- *Proofs.* The new construction and its quantitative estimates are proved in the text and appendices. The Boolean correspondence is invoked from [12, Theorem 4.9], with its local-dimension precision ingredients recorded in Section 10. The conjugacy gadgets are credited in Section 3. Results used for comparison are cited.
- *Proof audit.* A written refutation-first audit read the construction and its preceding clock and guard proofs. It found no fatal step or required change to the defining relations. It expanded the polynomial translation bound, the absence of a product of digit-depth losses, the all-branch measurement estimate and the entropy decoder, and checked the finite models with a disclosed test program.
- *Computation.* The archived finite-model script builds the operators at $N = 4, 6, 8$ with a fixed one-counter recognizer that halts exactly on even inputs. It records every defining

relation, required commutation, domain and state condition, and the derived decoding and reset identities. Its defect tables are in `anc/audit/recorded/`. A separate script multiplies the two conjugacy gadgets on finite matrices. No inequality in these tests was violated. Finite checks support the proofs and do not replace them; the arbitrary-state lower bound rests on the written proof.

- *Independent proof check.* A second refutation-first proof check was made by a model of another developer. It wrote its findings on the construction before reading the audit above, and then compared the two. It found no false step and no counterexample. It identified three repairs, each with a proof, and all three are applied here. The orthogonality context of the compiler now gives the projection E its own Boolean variable, defined by an exact AND context (Section 3). The list of joint relations in Lemma 9.3 now states explicitly the commutators between different compulsory clocks and between conditional and fuel or time measurements, and bounds them through the raw conjugated digits. The translation bound (9.4) is now linear in the number of translations, and the compulsory measurements are fixed explicitly. The check also recorded the two structural facts in Remark 9.4. Its two numerical scripts, for the gadgets, the selector identities and the orthogonality repair, are in `anc/crosscheck/`.
- *Independent computation.* A separately written program, built from the construction without the archived script or its output, evaluates the finite models of Section 7 at $N = 4, 6, 8$. It stores each operator as a sum of weighted basis permutations and uses its own three-counter test program. It evaluates every defining relation of Sections 4 and 5, every required commutation and domain condition, and the identities (6.1): 911, 915 and 919 relations. It computes operator norms, by exact block decomposition, as well as flat norms. Every relation stated to be exact has defect zero. The same ten relations as in the archived script have nonzero defects. The nine that do not depend on the program have the same flat ratios, and the conditional-tail ratio agrees at $N = 4, 8$, where the two test programs have the same K_N. The operator-norm defects are $3 \cdot 4^{-N}$ for the clocks, $\sqrt{2}\,4^{-N}$ for the guarded increments, 4^{-N} for the carry step and at most $0.012 \cdot 4^{-N}$ for (5.15), and the dimension is exactly 2^{3N+k_N}. Its code and recorded output are in the repository directory `verify/`. Like the archived script, it tests the finite models, not the arbitrary-state lower bound or the compiled game.

A Lean formalization of the version 1.0.0 core is in progress and is not complete. No completed Lean verification is claimed for this paper. The proofs, computations and review are offered for checking as they stand.

Version 2.0.0. The new results came from the same process of model-assisted research and drafting. A refutation-first spectrum audit expanded the reader induction, zero-tail histories, moving-window count, residual-power precision, and the tracial moment and compression arguments. A separate refutation-first usable-entanglement audit checked the inactive-outcome protocol, its original-mixed-state bridge and its operational rate. An independent usable-entanglement check by a model of another developer re-derived those steps and supplied sharper constants. An independent spectrum check by a model of another developer also found no false step; its clarifications on early halting, monotone approximations, domain encoding, arithmetic and synchronous hardness are incorporated. These expansions are in Sections 11–14. The programs and their recorded runs are in `anc/spectrum/` and `anc/extraction/`. They test finite reader models, window counts, compression moments and Jensen, and the actual extraction protocol, including noncommuting controls and mixed inputs. They do not prove the lower bounds. The ancillary README records the commands, archived outputs and limitations of these checks.

Complementary construction. Section 15 was checked by a refutation-first audit and by an independent check by a model of another developer. The latter required one repair, the use of separate projection families for the two players, which is incorporated here. The checks covered the tester induction, its non-self-adjoint register initialization, corner approximation, simultaneous finite models and the address-wraparound gauge. The original-state bridge and extraction proof apply to the new presentation, yielding its usable-entanglement and heat laws in Theorem 15.7. The exact external MIP*=RE theorem statements are cited there. The ancillary README specifies the commands, archived evidence and scope of the complementary-construction checks.

Priority. For each headline result, bounded searches of arXiv and the general literature compared the statements with the closest work; the credits they found are given at the results. They found no exact preemption of the composed fixed-game exponent, which is not an exhaustive priority verdict. The search included dimension witnesses and hyperlinear profiles, noncomputable exponents, symbolic entropy and frequency realization, computability and asymptotic density, radical-inverse discrepancy, almost synchronous compression, tracial moment hierarchies, one-way hashing, bounded polynomial and Boolean encodings, Cantor adding machines, and computation in finite presentations. The conjugacy gadgets are credited to Fanizza and coauthors; the measurement information bound to Holevo and the random-access-code literature; the operator-assignment interpretation to the constraint-game literature; and the state-preserving Boolean correspondence to the cited companion work. The finite presentation, its simultaneous models, and the matched same-state estimates are the specific construction proved here. For version 2.0.0 the searches also covered realization of computability classes as resource exponents, the MIP*=RE witness and commuting-value literature, and synchronous approximation.

A Why exact periodic verification is insufficient

This appendix records two obstructions that explain the use of approximate clocks and vanishing guard amplitudes. Neither is a general impossibility theorem for finite presentations. For the distinction between locally invariant finite marginals and globally extendible measures, see Chazottes, Gambaudo, Hochman and Ugalde [3]; the compactness argument below uses all box sizes, not extension of one finite marginal.

Proposition A.1 (Periodic extrema)**.** *Let X be a nonempty shift of finite type on $\mathbb{Z}^r$ and f a rational local observable. Put $\alpha = \min_\mu \int f \, d\mu$, where μ ranges over invariant probability measures. If averages of periodic configurations approach α, then α is computable.*

Proof. On each finite box, consider probability distributions on locally admissible patterns and impose every translation-invariance equation whose two supports stay in the box. This is a finite rational polytope. Minimize the expected value of f at the origin, obtaining a computable rational lower bound ℓ_N. Restriction from larger boxes makes these bounds monotone. A convergent subsequence of minimizers has consistent marginals on every finite support, gives an invariant measure on X, and has expectation $\lim \ell_N$. Thus $\ell_N \uparrow \alpha$. Periodic configurations and their finite fundamental domains are enumerable; local validity and their rational averages are decidable. Their successive minimum averages give upper bounds converging to their infimum. By hypothesis that infimum is α. Search until the upper and lower bounds differ by the desired error. □

Hochman–Meyerovitch's noncomputable extremal frequencies [16] therefore cannot be combined with arbitrarily good exact periodic models of those minima. In the present

upper construction the workspace histories close as labels, but the complete clocks remain approximate. Proposition A.1 does not apply to those clocks.

Theorem A.2 (Extraction under a fixed rejection gap)**.** *Suppose* $\mathbb{C}^d = \bigoplus_{v\in\mathcal{V}} \mathcal{H}_v$*, with* $M = |\mathcal{V}|$ *nonzero sector projections* P_v*. Unitaries* $U_1, \dots, U_s$ *permute the sectors exactly and satisfy* $\|[U_i, \xi]\|_2 \le \delta$*. For forbidden sets* $B_j \subseteq \mathcal{V}$*,* $1 \le j \le r$*, suppose*

$$D_j^* D_j \ge \kappa P_{B_j}, \qquad |D_j|_\rho \le \delta, \qquad P_{B_j} = \sum_{v\in B_j} P_v,$$

where $\kappa > 0$*. Let* $F = \sum_v c_v P_v$*,* $0 \le c_v \le 1$*, and* $a = M\delta$*. If* $2(s + r/\kappa)a^2 < 1$*, a transition orbit avoiding all forbidden sets has average cost at most*

$$\mathrm{Tr}(\rho F) + \sqrt{2s}\, a + 2(s + r/\kappa)a^2. \tag{A.1}$$

Proof. Let $p_v = \mathrm{Tr}(\rho P_v)$ and $q_v = \sqrt{p_v}$. If π_i is the sector permutation of U_i, the reverse triangle inequality applied to the row blocks $P_v\xi$ and $P_v U_i \xi U_i^*$ gives

$$\sum_v (q_v - q_{\pi_i(v)})^2 \le \delta^2.$$

The union B of forbidden sets has $p(B) \le (r/\kappa)\delta^2$. For an orbit O, let $E_O = \sum_i \sum_{v\in O} (q_v - q_{\pi_i(v)})^2$, so $\sum_O E_O \le s\delta^2$. A simple path in the connected orbit graph has at most $|O| - 1$ edges; Cauchy–Schwarz gives

$$(q_v - q_w)^2 \le (|O| - 1)E_O.$$

Choose a bad vertex w_O in each invalid orbit. Then

$$\sum_{v\in O} q_v^2 \le 2M q_{w_O}^2 + 2M^2 E_O.$$

Summing shows that the mass b of invalid orbits is at most $2(s + r/\kappa)a^2$.

Average p uniformly within each orbit, obtaining $\bar{p}$, and put $u_v = \sqrt{\bar{p}_v}$. The variance identity and the path bound give

$$\sum_{v\in O} (q_v - \bar{q}_O)^2 \le M^2 E_O,$$

where $\bar{q}_O$ is the ordinary average of q on O. If the orbit has mass $p_O > 0$, set $t_O = (\sum_{v\in O} q_v)/\sqrt{|O| p_O}$. Then

$$\sum_{v\in O} (q_v - u_v)^2 = 2p_O(1 - t_O) \le 2p_O(1 - t_O^2) = 2\sum_{v\in O} (q_v - \bar{q}_O)^2.$$

Consequently $\|q - u\|_2 \le \sqrt{2s}\, M\delta$. Since $\mathrm{TV}(p, \bar{p}) \le \frac{1}{2}\|q - u\|_2 \|q + u\|_2 \le \|q - u\|_2$, orbit averaging changes expected cost by at most this amount. Conditioning $\bar{p}$ on valid orbits changes total variation by $b < 1$. The result is a mixture of uniform valid-orbit laws, one of which has cost at most (A.1). □

Corollary A.3. *Assume valid finite orbits are effectively enumerable, have computable rational costs, and every such cost is at least a left-c.e. real* θ*. If near-minimizing models have flag cost tending to* θ *and* $M\delta \to 0$*, then* θ *is computable. If* θ *is uncomputable and flat models have flag cost tending to* θ*, their dimensions eventually satisfy* $d \ge \kappa\delta^{-2}$*.*

Proof. Theorem A.2 makes the infimum of valid finite-orbit costs equal to θ under the first hypothesis. Enumerating those costs gives computable upper bounds tending to θ; left semicomputability gives lower bounds. This computes θ.

More generally, for uncomputable θ the infimum θ_{fin} of valid finite-orbit costs must be strictly greater than θ, or there are no valid finite orbits: equality would already make θ computable by the same enumeration. If a flat model had no nonzero forbidden projection, every orbit would be valid. Exact sector permutation makes ranks constant on an orbit, so its flat cost would be a mixture of uniform orbit costs, at least θ_{fin}. Thus near-minimizing flat models eventually have a nonzero P_{B_j}. The rejection gap gives

$$\delta^2 \geq d^{-1} \operatorname{Tr}(D_j^* D_j) \geq \kappa/d.$$

This proves the claimed dimension bound. □

The semantic assumption about all valid finite-orbit costs is essential. A program orbit may fail to extend to the full clock system, or may have a cheap flag cost that the extra clock relations exclude. The corollary does not apply automatically to every operator presentation. Its flat bound explains the amplitude issue: the target dimensions here are below order δ^{-2}, so a construction satisfying these hypotheses could not use fixed-gap bad sectors. Our increment defect instead has amplitude 4^{-N} at overflow, as Proposition 4.5 proves. The initialization proof uses exact tracial cancellation and compactness, not a fixed-gap finite-sector extraction.

B Finite computations and reproducibility

The code in `anc/audit/check_models.py` represents an operator as a finite sum of weighted basis permutations. For each defining relation it enumerates all basis columns of the tensor factors that occur in that relation. The absent factors are identities, so averaging squared column norms computes exactly the full-space normalized Hilbert–Schmidt expression, before floating-point rounding. Equal output rows are combined before squaring, avoiding cancellation of large squared terms. The periodic coordinates in (7.6) are summed with integer numerators and denominator $4^{\ell} - 1$ for the permutation period ℓ, then rounded once to floating point.

The test program has one counter and five controls. From the starting control, zero halts, while nonzero decrements and moves to an odd-parity control. There zero enters an absorbing nonhalting control, while nonzero decrements and returns to start. Idle and halt are absorbing. Hence input e halts exactly when it is even, at step $e + 1$. This is a test of the finite-model mechanism, not a computation of the universal halting set.

The remaining defining defects are zero to within 10^{-14} in these tests. Complete per-relation values and their reference scales are archived, not just the maxima in Table 1. The two guarded increments have identical ratios in this example. The theorem's operator-norm bounds are proved in Section 7; a script's maximum column norm is not generally an operator norm. The displayed dimensions are represented without dense matrices. The original $N = 8$ run had an observed peak working set about 1.16 GB.

The gadget script `anc/gadgets/gadget_check.py` tests self-adjointness, squares, all pairwise commutations within each context, context products, and the auxiliary conjugacies on random finite instances. It is a numerical check of the explicit extension, not a proof of the converse; that converse is proved in Section 3.

The ancillary README gives commands, dependencies, recorded outputs and limitations. All computations are run with one numerical-library thread. The finite-model script caps its exhaustive workspace implementation at $N = 8$ to stay within the stated memory budget. The independent engine described in the methods section is in the repository directory

Table 1: Finite-model checks for the disclosed test program. Each ratio is a flat Hilbert–Schmidt defect divided by 4^{-N}. The first two rows apply separately to all three compulsory clocks.

Quantity	$N = 4$	$N = 6$	$N = 8$
Clock affine ratio	$3/\sqrt{2}$	$3/\sqrt{2}$	$3/\sqrt{2}$
Clock tail-flip ratio	3	3	3
Guarded increment ratio	0.353553	0.176777	0.088388
Carry-step ratio	0.25	0.125	0.0625
Conditional tail ratio	0.00585946	0.000160228	0.00256366
K_N	$\{0\}$	$\{0, 2\}$	$\{0, 2\}$
k_N	2	4	4
Dimension	16 384	4 194 304	268 435 456
Recorded checks	390	394	398

`verify/`, with its recorded output, and the two scripts of the independent proof check are in `anc/crosscheck/`.

Version 2 checks. The spectrum program in `anc/spectrum/` builds 24 finite reader models at $N = 2, 4, 6$ and eight rational thresholds, including both endpoints. It tests reset boundaries, reads beyond N, halt inclusion, slot collisions, dressing and defining operator residuals on the nontrivial kernels. Its other tests cover radical-inverse windows, compression with domain repairs, finite word moments and noncommuting-state Jensen. The protocol programs in `anc/extraction/` evaluate the stage inequality, full exact and perturbed protocols with mixed inputs and noncommuting controls, small perturbations with positive lower bounds, bounded adversarial searches, and one-sided inclusion repair. Their recorded outputs are preserved in the respective `recorded/` directories. The ancillary README states each file's scope and distinguishes the original allowances from the constants of Lemma 12.1.

Complementary-construction checks. The script `anc/gate/toy_gate.py` retains its original recorded results in `gate/recorded/` and writes a new result beside the script. It executes the signed packed polynomial instructions for three-colouring of K_3, a pair of noncommuting binary bases, and a flagged bad three-colouring of K_4. It checks contraction-register decompositions, initialized finite histories through every distinct reset period, cyclic boundaries, simultaneous data factors, decoder and reset recursions, complementary window counts and address dressing. This script was written for an earlier presentation with a single projection family for both players; its history, reset and window tests do not depend on that choice and cover 16,893 history transitions, 35 cyclic reset boundaries and 2,396,744 mask/start cases across 254 masks. The separate two-family suite in `anc/gate/crosscheck/` tests the presentation of Section 15.1, with separate families for the two players: K_3 colouring has 18 generators, and the magic-square witness has 42 generators and packed dimension 168. It also checks the reduced-state eigenspaces, an anti-synchronous PCC control, 53,248 history points, sparse address models and windows through depth 22. Original logs and rerun commands are specified in the ancillary README. These are tests of the new matrix algebra and finite histories on known-answer toy games, not the MIP* games themselves. They do not certify the halting reduction, arbitrary-state lower bound or compiled game.

References

[1] Johannes Bausch, Toby S. Cubitt, and James D. Watson. Uncomputability of phase diagrams. *Nature Communications*, 12:452, 2021. arXiv:1910.01631, doi:10.1038/s41467-020-20504-6.

[2] Sabine Burgdorf and Igor Klep. The truncated tracial moment problem. *Journal of Operator Theory*, 68(1):141–163, 2012. arXiv:1001.3679.

[3] Jean-René Chazottes, Jean-Marc Gambaudo, Michael Hochman, and Edgardo Ugalde. On the finite-dimensional marginals of shift-invariant measures. *Ergodic Theory and Dynamical Systems*, 32(5):1485–1500, 2012. arXiv:1011.2442, doi:10.1017/S0143385711000526.

[4] Richard Cleve, Li Liu, and William Slofstra. Perfect commuting-operator strategies for linear system games. *J. Math. Phys.*, 58(1):012202, 2017. arXiv:1606.02278, doi:10.1063/1.4973422.

[5] Richard Cleve and Rajat Mittal. Characterization of binary constraint system games. In *Automata, Languages, and Programming (ICALP 2014)*, volume 8572 of *Lecture Notes in Comput. Sci.*, pages 320–331. Springer, 2014. arXiv:1209.2729, doi:10.1007/978-3-662-43948-7_27.

[6] Andrea Coladangelo. A two-player dimension witness based on embezzlement, and an elementary proof of the non-closure of the set of quantum correlations. *Quantum*, 4:282, 2020. arXiv:1904.02350, doi:10.22331/q-2020-06-18-282.

[7] Alain Connes. Classification of injective factors. cases II_1, II_∞, III_λ, $\lambda \neq 1$. *Ann. of Math. (2)*, 104(1):73–115, 1976. doi:10.2307/1971057.

[8] Toby S. Cubitt, David Perez-Garcia, and Michael M. Wolf. Undecidability of the spectral gap. *Forum of Mathematics, Pi*, 10:e14, 2022. arXiv:1502.04573, doi:10.1017/fmp.2021.15.

[9] Igor Devetak and Andreas Winter. Distillation of secret key and entanglement from quantum states. *Proceedings of the Royal Society A*, 461(2053):207–235, 2005. The hashing inequality is Theorem 10 of the arXiv version. arXiv:quant-ph/0306078, doi:10.1098/rspa.2004.1372.

[10] Seth Douglas and Nidhal Mghirbi. The heat price of precision, 2026. Version 1.0.0. Unrefereed preprint; theorem numbers refer to this version. doi:10.5281/zenodo.23701732.

[11] Seth Douglas and Nidhal Mghirbi. A nonlocal game whose dimension exponent is uncomputable, 2026. Version 1.0.0. Concept DOI: 10.5281/zenodo.23701733. Unrefereed preprint. doi:10.5281/zenodo.23701734.

[12] Seth Douglas and Nidhal Mghirbi. The price of precision: usable entanglement in nonlocal games and the geometry of groups, 2026. Version 1.0.0. Unrefereed preprint; theorem numbers refer to this version. doi:10.5281/zenodo.23697346.

[13] Rodney G. Downey, Carl G. Jockusch, Jr., and Paul E. Schupp. Asymptotic density and computably enumerable sets. *Journal of Mathematical Logic*, 13(2):1350005, 2013. arXiv:1307.0040, doi:10.1142/S0219061313500050.

[14] Marco Fanizza, Larissa Kroell, Arthur Mehta, Connor Paddock, Denis Rochette, William Slofstra, and Yuming Zhao. The NPA hierarchy does not always attain the commuting operator value, 2025. Result numbers refer to arXiv version 4. arXiv:2510.04943.

[15] Andrew Hawkins, Adam Skalski, Stuart White, and Joachim Zacharias. On spectral triples on crossed products arising from equicontinuous actions. *Mathematica Scandinavica*, 113(2):262–291, 2013. arXiv:1103.6199, doi:10.7146/math.scand.a-15572.

[16] Michael Hochman and Tom Meyerovitch. A characterization of the entropies of multidimensional shifts of finite type. *Ann. of Math. (2)*, 171(3):2011–2038, 2010. doi:10.4007/annals.2010.171.2011.

[17] Alexander S. Holevo. Bounds for the quantity of information transmitted by a quantum communication channel. *Probl. Inf. Transm.*, 9(3):177–183, 1973. Translated from Problemy Peredachi Informatsii 9(3), 3–11.

[18] Zhengfeng Ji, Debbie Leung, and Thomas Vidick. A three-player coherent state embezzlement game. *Quantum*, 4:349, 2020. arXiv:1802.04926, doi:10.22331/q-2020-10-26-349.

[19] Zhengfeng Ji, Anand Natarajan, Thomas Vidick, John Wright, and Henry Yuen. MIP*=RE, 2026. arXiv version 4, accepted to Annals of Mathematics. Result and page numbers refer to this version. URL: https://arxiv.org/abs/2001.04383v4, arXiv:2001.04383.

[20] Carl G. Jockusch, Jr. and Paul E. Schupp. Generic computability, Turing degrees, and asymptotic density. *Journal of the London Mathematical Society*, 85(2):472–490, 2012. arXiv:1010.5212, doi:10.1112/jlms/jdr051.

[21] Se-Jin Kim, Vern I. Paulsen, and Christopher Schafhauser. A synchronous game for binary constraint systems. *Journal of Mathematical Physics*, 59(3):032201, 2018. arXiv:1707.01016, doi:10.1063/1.4996867.

[22] Igor Klep, Victor Magron, and Jurij Volčič. Optimization over trace polynomials. *Annales Henri Poincaré*, 23:67–100, 2022. arXiv:2006.12510, doi:10.1007/s00023-021-01095-4.

[23] Artemy Kolchinsky and David H. Wolpert. Thermodynamic costs of Turing machines. *Physical Review Research*, 2:033312, 2020. arXiv:1912.04685, doi:10.1103/PhysRevResearch.2.033312.

[24] Junqiao Lin. MIP^{co}=coRE, 2025. Commuting-operator interactive proofs; arXiv version 1. URL: https://arxiv.org/abs/2510.07162, arXiv:2510.07162.

[25] Arthur Mehta, William Slofstra, and Yuming Zhao. Positivity is undecidable in tensor products of free algebras. *J. Eur. Math. Soc. (JEMS)*, 2026. Published online. Result numbers refer to arXiv version 2. arXiv:2312.05617, doi:10.4171/JEMS/1786.

[26] Nidhal Mghirbi and Seth Douglas. An explicit nonlocal game of quantum value one that needs exponential entanglement, 2026. Version 1.0.0, Zenodo. Unrefereed preprint. doi:10.5281/zenodo.23697255.

[27] Hamoon Mousavi, Seyed Sajjad Nezhadi, and Henry Yuen. On the complexity of zero gap MIP*. In *47th International Colloquium on Automata, Languages, and Programming (ICALP 2020)*, volume 168 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 87:1–87:12, 2020. URL: https://arxiv.org/abs/2002.10490, arXiv:2002.10490, doi:10.4230/LIPIcs.ICALP.2020.87.

[28] Hamoon Mousavi, Seyed Sajjad Nezhadi, and Henry Yuen. Nonlocal games, compression theorems, and the arithmetical hierarchy. In *Proceedings of the 54th Annual ACM SIGACT Symposium on Theory of Computing (STOC 2022)*, pages 1–11. ACM, 2022. Result numbers refer to arXiv version 2. arXiv:2110.04651, doi:10.1145/3519935.3519949.

[29] Miguel Navascués, Stefano Pironio, and Antonio Acín. A convergent hierarchy of semidefinite programs characterizing the set of quantum correlations. *New Journal of Physics*, 10:073013, 2008. arXiv:0803.4290, doi:10.1088/1367-2630/10/7/073013.

[30] Ashwin Nayak. Optimal lower bounds for quantum automata and random access codes. In *Proceedings of the 40th Annual Symposium on Foundations of Computer Science*, pages 369–376. IEEE, 1999. arXiv:quant-ph/9904093, doi:10.1109/SFFCS.1999.814608.

[31] Vern I. Paulsen, Simone Severini, Daniel Stahlke, Ivan G. Todorov, and Andreas Winter. Estimating quantum chromatic numbers. *Journal of Functional Analysis*, 270(6):2188–2222, 2016. arXiv:1407.6918, doi:10.1016/j.jfa.2016.01.010.

[32] Mark V. Sapir, Jean-Camille Birget, and Eliyahu Rips. Isoperimetric and isodiametric functions of groups. *Annals of Mathematics*, 156(2):345–466, 2002. arXiv:math/9811105, doi:10.2307/3597195.

[33] J. C. Shepherdson and H. E. Sturgis. Computability of recursive functions. *Journal of the ACM*, 10(2):217–255, 1963. doi:10.1145/321160.321170.

[34] William Slofstra. A group with at least subexponential hyperlinear profile, 2018. arXiv:1806.05267.

[35] William Slofstra. The set of quantum correlations is not closed. *Forum Math. Pi*, 7:e1, 2019. Result numbers refer to arXiv version 2. arXiv:1703.08618, doi:10.1017/fmp.2018.3.

[36] William Slofstra. Tsirelson's problem and an embedding theorem for groups arising from non-local games. *J. Amer. Math. Soc.*, 33(1):1–56, 2020. arXiv:1606.03140, doi:10.1090/jams/929.

[37] William Slofstra and Thomas Vidick. Entanglement in non-local games and the hyperlinear profile of groups. *Ann. Henri Poincaré*, 19(10):2979–3005, 2018. Result numbers refer to arXiv version 2. arXiv:1711.10676, doi:10.1007/s00023-018-0718-y.

[38] Ernst Specker. Nicht konstruktiv beweisbare Sätze der Analysis. *Journal of Symbolic Logic*, 14(3):145–158, 1949. doi:10.2307/2267043.

[39] Johannes G. van der Corput. Verteilungsfunktionen. I. *Proceedings of the Koninklijke Nederlandse Akademie van Wetenschappen*, 38:813–821, 1935. Binary radical-inverse construction; the window estimate used here is proved in the text.

[40] Thomas Vidick. Almost synchronous quantum correlations. *Journal of Mathematical Physics*, 63(2):022201, 2022. arXiv:2103.02468, doi:10.1063/5.0056512.